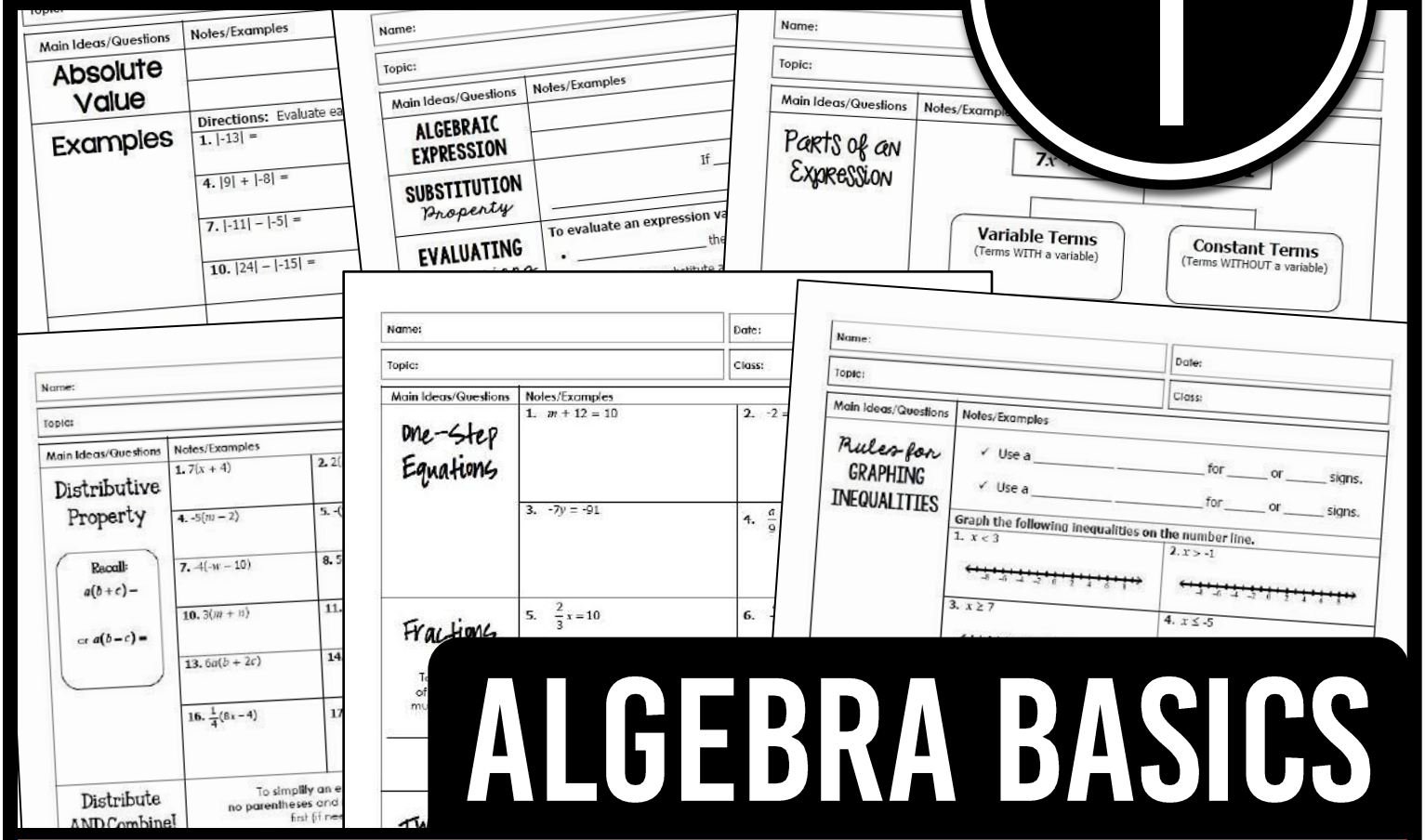
ALGEBRA I

Unit 1



ALGEBRA BASICS

NOTES • HOMEWORK • QUIZZES • TEST

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Unit 1 – Algebra Basics: Sample Unit Outline

	TOPIC	HOMEWORK
DAY 1	The Real Number System	HW #1
DAY 2	Properties	HW #2
DAY 3	"Sell that Property" Performance Task	HW #3
DAY 4	**Quiz 1-1**	None
DAY 5	Order of Operations & Absolute Value	HW #4
DAY 6	Evaluating Expressions	HW #5
DAY 7	Matrices	HW #6
DAY 8	**Quiz 1-2**	None
DAY 9	Combining Like Terms	HW #7
DAY 10	Simplifying Expressions (Distribute/Combine)	HW #8
DAY 11	Translating Expressions/Equations/Inequalities	HW #9
DAY 12	**Quiz 1-3**	None
DAY 13	Solving Two-Step Equations	HW #10
DAY 14	Solving & Graphing Two-Step Inequalities	HW #11
DAY 15	"Translating and Solve" Practice Worksheet Unit 1 Test Review	Complete Study Guide
DAY 16	**UNIT 1 TEST**	None

See sample images of the pages on the next page.

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The Real Number System

REAL NUMBERS

_____ = _____

Decimals that never end and have no repeating pattern. *Think ***MUMBO JUMBO NUMBERS***

IRRATIONAL NUMBERS

_____ = _____

All fractions and decimals that end or have a repeated pattern.

RATIONAL NUMBERS

_____ = _____

INTEGERS

_____ = _____

WHOLE NUMBERS

_____ = _____

NATURAL NUMBERS

_____ = _____

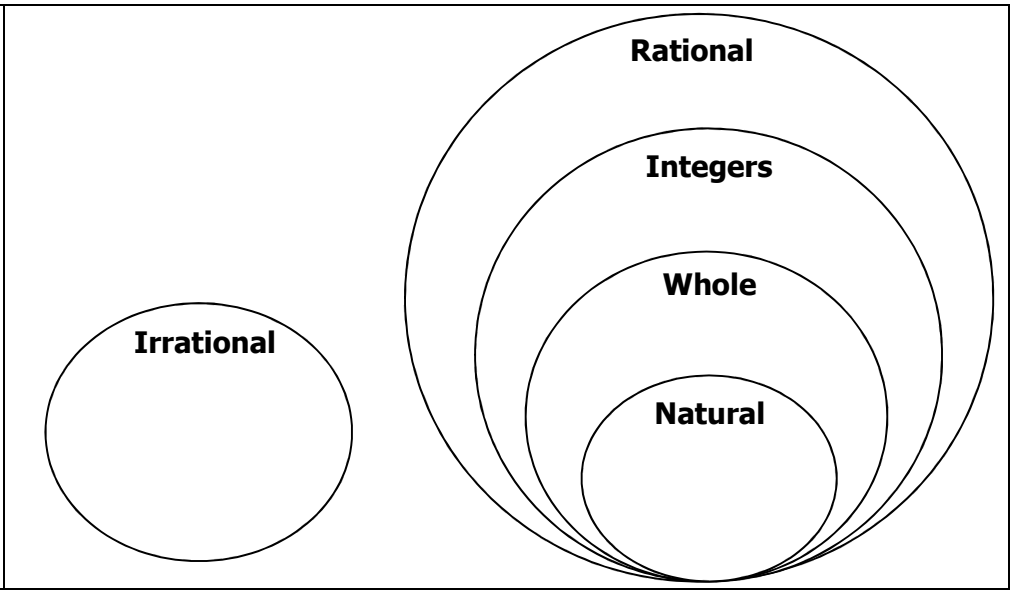
Directions: Name all sets of numbers to which each number belongs.		
1. 30	**2.** -11	**3.** $5\frac{4}{7}$
4. $\sqrt{21}$	**5.** 0	**6.** $-\sqrt{9}$
7. $\frac{6}{3}$	**8.** π	**9.** $5.\overline{3}$
10. $\frac{\sqrt{144}}{-3}$	**11.** -2.875	**12.** $\frac{\sqrt{17}}{\sqrt{17}}$

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Organizing the Real Numbers

Directions: Place the LETTER of each value in its location in the real number system below.

A. 7	**B.** $-\frac{36}{12}$
C. $-\sqrt{36}$	**D.** π
E. 6.48	**F.** -2
G. $-\sqrt{10}$	**H.** 0
I. $2\frac{1}{6}$	**J.** $-0.\overline{7}$
K. $\sqrt{\frac{4}{81}}$	



1. Which expression does **not** name an integer?

A. -15 **C.** 0

B. $\frac{6}{14}$ **D.** $\frac{12}{6}$

2. Which expression represents an irrational number?

A. $0.\overline{18}$ **C.** $\frac{2}{3}$

B. $\sqrt{75}$ **D.** $\sqrt{3}-\sqrt{3}$

3. Which number is **not** a whole number?

A. 8 **C.** 0

B. -10 **D.** $\frac{18}{3}$

4. Which of the following is a true statement?

A. -9 is a whole number

B. $\sqrt{25}$ is an irrational number

C. 0 is a natural number

D. $\frac{2}{3}$ is a rational number

5. Which of the following statements is false?

A. All real numbers are rational numbers.

B. Every integer is a rational number.

C. All natural numbers are integers.

D. Every whole number is a real number.

6. Give an example of a number that is a whole number, but not a natural number. _____

7. Give an example of a real number that is not rational. _____

8. Give an example of a rational number that is not an integer. _____

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Name: _____ **Unit 1:** Algebra Basics

Date: _____ **Bell:** _____ **Homework 1:** The Real Numbers

Directions: Name all sets of numbers to which each real number belongs.

1. 12

2. -15

3. $1\frac{1}{2}$

4. 3.18

5. $\sqrt{48}$

6. $9.\overline{3}$

7. $-2\frac{7}{9}$

8. $\sqrt{25}$

9. $\sqrt{3}$

10. $-\sqrt{64}$

11. $-\sqrt{12}$

12. $\frac{8}{4}$

Directions: Place the LETTER of each value in its location in the real number system below.

A. 2.125

B. 0

C. π

D. $-\frac{15}{5}$

E. $3\frac{1}{4}$

F. $0.\overline{4}$

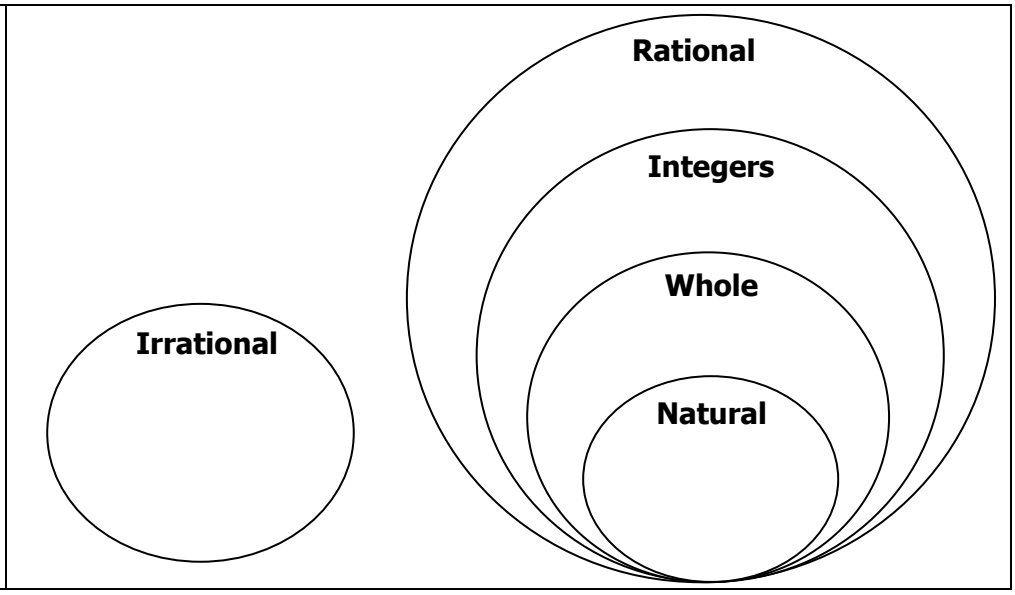
G. 23

H. -1

I. $\sqrt{\frac{16}{25}}$

J. $-\sqrt{49}$

K. $-\sqrt{35}$



Questions:

1. Give an example of a whole number that is not a natural number. _____

2. Give an example of a rational number that is not a whole number. _____

3. Give an example of a real number that is irrational. _____

4. Are integers always/sometimes/never whole numbers? _____

5. Are natural numbers always/sometimes/never rational numbers? _____

6. Are irrational numbers always/sometimes/never rational numbers? _____

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PROPERTIES OF REAL NUMBERS

In math, properties are statements that are true for any numbers.
They justify steps when simplifying expressions and solving equations.

PROPERTY	MAIN IDEA	EXAMPLES
COMMUTATIVE *Property* (of Addition or Multiplication)	_____ of values does not matter!	
ASSOCIATIVE *Property* (of Addition or Multiplication)	_____ of values does not matter!	
IDENTITY *Property* (of Addition or Multiplication)	_____ _____ _____!	
INVERSE *Property* (of Addition or Multiplication)	Using _____ to "cancel" a value!	
PROPERTY OF ZERO	Multiplying by _____ always equals _____!	
DISTRIBUTIVE *Property*	_____ a value to an expression inside _____.	

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New PROPERTIES

PROPERTY	MAIN IDEA	EXAMPLES
REFLEXIVE *Property*		
SYMMETRIC *Property*		
TRANSITIVE *Property*		

CLOSURE *Property*	A set is **closed** (under an operation) if the operation always produces an element of the same set. If an element outside the set is produced, then the operation is **not closed.**
	Directions: Answer True/False. If false, give a counterexample.
	1) Integers are closed under multiplication. _____ **2)** Natural numbers are closed under subtraction. _____ **3)** Even numbers are closed under addition. _____ **4)** Irrational numbers are closed under division. _____ **5)** Perfect squares are closed under subtraction. _____

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Name: _____ **Unit 1:** Algebra Basics

Date: _____ **Bell:** _____ **Homework 2:** Algebraic Properties

**** This is a 2-page document! ****

1. Commutative Property

Explain: _____

_____

_____

Two Examples (NOT from class!)

- _____
- _____

2. Associative Property

Explain: _____

_____

_____

Two Examples (NOT from class!)

- _____
- _____

3. Identity Property

Explain: _____

_____

_____

Two Examples (NOT from class!)

- _____
- _____

4. Inverse Property

Explain: _____

_____

_____

Two Examples (NOT from class!)

- _____
- _____

5. Property of Zero

Explain: _____

_____

_____

Two Examples (NOT from class!)

- _____
- _____

6. Distributive Property

Explain: _____

_____

_____

Two Examples (NOT from class!)

- _____
- _____

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7. Reflexive Property

Explain: _____

_____

_____

Two Examples (NOT from class!)

- _____
- _____

8. Symmetric Property

Explain: _____

_____

_____

Two Examples (NOT from class!)

- _____
- _____

9. Transitive Property

Explain: _____

_____

_____

One Example (NOT from class!)

- _____
- _____

10. Use #1-9 to identify each property below.

_____ **A.** $ab^2 \cdot 0 = 0$

_____ **B.** (7 + 5) + 1 = 7 + (5 + 1)

_____ **C.** 6 = 6

_____ **D.** $7(x - 3) = 7x - 21$

_____ **E.** $x + (-x) = 0$

_____ **F.** if $y = 3$, then $3 = y$

_____ **G.** $(a + b) + 0 = 0 + (a + b)$

_____ **H.** if $x = -1$, and $-1 = z$, then $x = z$

_____ **I.** $4x \cdot 1 = 4x$

11. Is the set of integers closed under division? **Explain why or why not.**
If it is not closed, give an example.

12. Is the set of real numbers closed under addition? **Explain why or why not.**
If it is not closed, give an example.

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Properties Warm-Up!!!

_____ **1.** 16 = 16

_____ **2.** $7 + (a + b) = (7 + a) + b$

_____ **3.** $(4y + 1) \cdot 0 = 0$

_____ **4.** If $5 \cdot 2 = 10$ and $10 = \sqrt{100}$, then $5 \cdot 2 = \sqrt{100}$

_____ **5.** $-2(x - y) = -2x + 2y$

_____ **6.** $3 \cdot \frac{1}{3} = 1$

_____ **7.** If $8x = 24$, then $24 = 8x$

_____ **8.** $(2 + 1) + 9 = (1 + 2) + 9$

_____ **9.** $6m \cdot 1 = 6m$

A. Commutative Property

B. Associative Property

C. Distributive Property

D. Identity Property

E. Inverse Property

F. Property of Zero

G. Reflexive Property

H. Symmetric Property

I. Transitive Property

10. True/False: Negative numbers closed under subtraction. Provide an example if false.

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Sell That Property!



Unit 1 Performance Task

Properties of Real Numbers

Currently, math students are struggling to understand the properties of real numbers. This is having an impact on their ability to solve problems. The Math Real Estate Group has hired you to create a poster showing the applications of the real number properties. You will be working in teams and each given one property to display. Your goal is to create a **slogan** for the property, an **explanation** of what the property does, **visuals**, and **at least 4 examples** of the property in action. You want to encourage other students to "buy" your property!

This poster must be creative, unique, and show your understanding of the property.

Task Rubric:	**4** Real Understanding!	**3** Real Close!	**2** "Real" It in!	**1** Not Really!
Mathematical Concepts (Explanation of Property)	Explanation is detailed and clear. Correct terminology and notation are always used.	Explanation is clear. Correct terminology and notation are almost always used.	Explanation is a little difficult to understand. Correct terminology is sometimes used.	Explanation is difficult to understand or is missing. There is little use, or a lot of inappropriate use, of terminology and notation.
Effectiveness	Outstanding use of slogan and visuals to depict importance of property. High degree of effectiveness.	Good use of slogan and visuals to depict importance of property.	Slogans and visuals somewhat depict importance of property.	Slogans and visuals incorrectly depict importance of property OR were not included.
Examples	All example problems are completed and correct.	All but one of the example problems are completed and correct.	All but two of the example problems are completed and correct.	Three or more of the example problems are not completed or incorrect.
Neatness and Organization	The work is presented in a neat, clear, organized fashion that is easy to read.	The work is presented in a neat and organized fashion that is usually easy to read.	The work is presented in an organized fashion but may be hard to read at times.	The work appears sloppy and unorganized. It is hard to know what information goes together.

Project Grade: _____

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Name: _____ **Unit 1:** Algebra Basics

Date: _____ **Bell:** _____ **Homework 3:** Quiz 1 Review

Directions: Name all sets of numbers to which each real number belongs.		
1. $-\frac{14}{2}$	**2.** $\sqrt{64}$	**3.** 0
4. π	**5.** $0.\overline{45}$	**6.** $\frac{3}{8}$

Directions: Place the LETTER of each value in its location in the real number system below.

A. $-0.\overline{2}$ **B.** 18

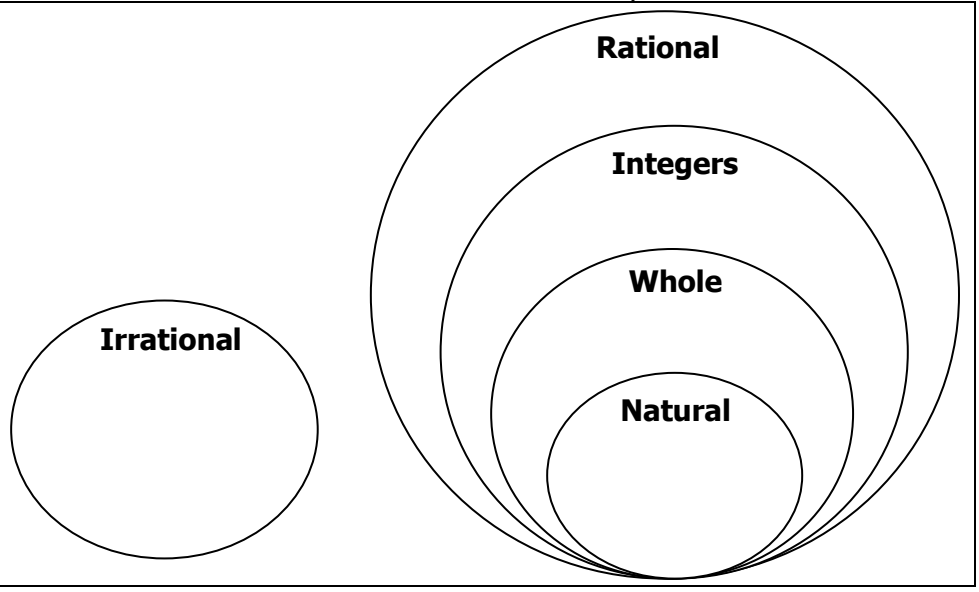
C. $-\sqrt{100}$ **D.** π

E. 0 **F.** $2\frac{1}{6}$

G. -5 **H.** 4.03

I. $-\sqrt{50}$ **J.** $\sqrt{\frac{4}{9}}$

K. $\frac{36}{9}$



Directions: Name the property shown.

1. $4 + (x + y) = (4 + x) + y$	**6.** $2\left(\frac{1}{2}\right) = 1$
2. if $\sqrt{49} = 7$ and $7 = 3 + 4$, then $\sqrt{49} = 3 + 4$	**7.** $3 = 3$
3. $(mn)\cdot 1 = mn$	**8.** $10k^2 + (-10k^2) = 0$
4. $(x+3)\cdot 0 = 0$	**9.** $5(x + 7) = 5x + 35$
5. $(x + 2) + y = (2 + x) + y$	**10.** if $-7v = 28$, then $28 = -7v$

***Also, make sure you know the closure property!**

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Name: _____ **Algebra I**

Date: _____ **Bell:** _____ **Unit 1:** Algebra Basics

Quiz 1-1: The Real Numbers & Properties

For Questions 1 – 5: Put an X in each set the number belongs.

	R (Real)	**I** (Irrational)	**Q** (Rational)	**Z** (Integers)	**W** (Whole)	**N** (Natural)
1. 10						
2. $\frac{5}{6}$						
3. $\sqrt{24}$						
4. 0						
5. $-\sqrt{81}$						

6. Place the LETTER of each value in its location in the real number system below.

A. $\sqrt{0}$ **B.** $\sqrt{48}$

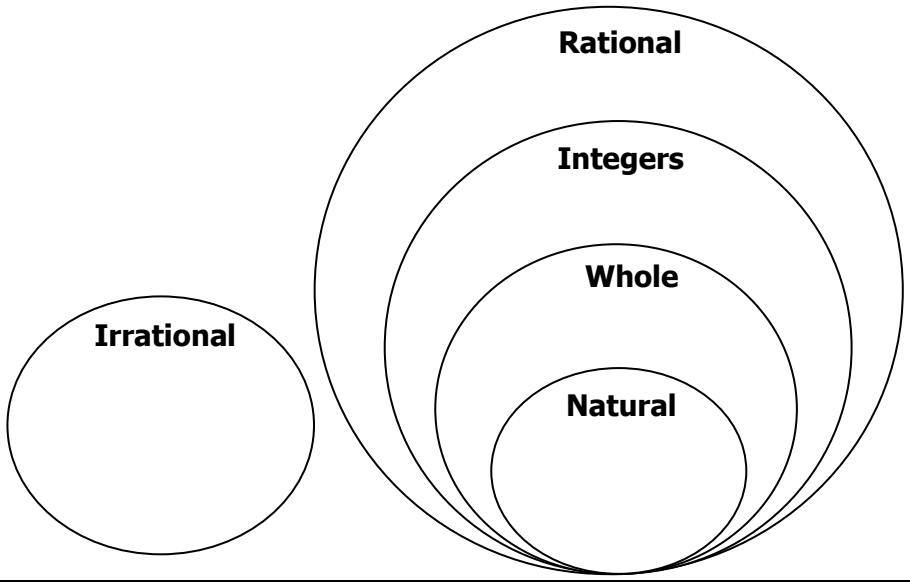
C. $-\frac{24}{8}$ **D.** 17

E. π **F.** -3.75

G. -1 **H.** $\sqrt{36}$

I. $\sqrt{\frac{16}{49}}$ **J.** $0.\overline{9}$

K. $2\frac{1}{5}$



Directions: Circle the best answer.

7. Which does **not** represent an integer? **A.** 0 **B.** $-\frac{18}{2}$ **C.** $\sqrt{9}$ **D.** $\frac{4}{5}$

8. Which is an irrational number? **A.** $\sqrt{21}$ **B.** $3\frac{5}{8}$ **C.** $2.\overline{41}$ **D.** -13

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9. Which statement is true?

A. $\sqrt{81}$ is an irrational number

B. 0 is a natural number

C. $4.\overline{6}$ is a rational number

D. -9 is a whole number

10. Which statement is false?

A. All whole numbers are integers.

B. All real numbers are irrational numbers.

C. All natural numbers are whole numbers.

D. All whole numbers are real numbers.

For Questions 11 – 18: Identify the property that justifies the statement.
Use the property bank below for spelling!

11. If y = -5, then -5 = y	**12.** $8a + 4 = 4 + 8a$
13. $12(x + 4) = 12x + 48$	**14.** $n^2 \cdot 0 = 0$
15. $8 + (2 + 6) = (8 + 2) + 6$	**16.** If $3^2 = 9$ and $9 = \sqrt{81}$, then $3^2 = \sqrt{81}$
17. -7 = -7	**18.** $6s \cdot \frac{1}{6s} = 1$

Property Bank:

Commutative
Associative
Distributive
Identity
Inverse
Property of Zero
Reflexive
Symmetric
Transitive

19. Which set of numbers is closed under subtraction?

A. Whole Numbers

B. Natural Numbers

C. Integers

D. Irrational Numbers

20. Which of the following sets is NOT closed under addition?

A. Natural Numbers

B. Whole Numbers

C. Integers

D. Odd Numbers

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Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples
Absolute Value	
Examples	**Directions:** Evaluate each expression.

1. |-13| =	**2.** |21| =	**3.** |-3| + |-5| =
4. |9| + |-8| =	**5.** |-12| + |15| =	**6.** |21 – 17| =
7. |-11| – |-5| =	**8.** |4| – |-4| =	**9.** |15 – 23| =
10. |24| – |-15| =	**11.** |28| – |-26| =	**12.** |-3| – |19| =

Order of Operations	

Examples	**Directions:** Evaluate each expression.

13. $7+54\div3(2)$	**14.** $17-2^5\div4+6$
15. $24-4^2\cdot3+15$	**16.** $\dfrac{(7-2^2)+17}{-14+2\cdot5}$

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	17. $(8-5)^3-\lvert 5^2-4^3\rvert \div 3$	**18.** $\left\lvert -7^2+3^2\cdot 4\right\rvert +18\div 2\cdot 5$
	19. $\dfrac{5^2-3^4}{(2^3-5)\cdot(6^2\div 9)}$	**20.** $-\left\lvert -9-10\right\rvert +12^2\div 6\cdot 3$
	21. $20^2\div\{3(5-9)^2+2\}$	**22.** $\dfrac{12\left[30-(9+4^2)\right]}{\lvert 10\rvert-\lvert -6\rvert}$
	23. $\dfrac{84}{12}+(5-7^2)-72\div 6\cdot 2$	**24.** $\dfrac{-(8-11)^3-3\lvert -14\rvert}{2^4-7}$

Summary: _____

_____

_____

_____

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Name: _____ **Unit 1:** Algebra Basics

Date: _____ **Bell:** _____ **Homework 4:** Order of Operations

Directions: Evaluate each expression using the order of operations.	
1. $17 - 5 \cdot 4 \div 2$	**2.** $40 - 32 \div 8 + 5 \cdot -2$
3. $35 - 14 \div 2 + 8^2$	**4.** $(4-7)^2 - 6 \cdot 7 + 20$
5. $2[18 - (5 + 3^2) \div 7]$	**6.** $[(-5+1) \div 2]^3 - \lvert -7 \rvert$
7. $1 + (-2-5)^2 + (14-17) \cdot 4$	**8.** $\frac{6(2+4)-1}{2 \cdot 3 + 1}$
9. $\frac{2^5 \div 8 + 3}{4+3}$	**10.** $\frac{17 \cdot 5 - 3 \cdot 5}{3^2 + 1}$
11. $\frac{13 + 7^2 \div 7}{9 - 20 \div 4 - 16}$	**12.** $-\lvert -16 \rvert \div \frac{(3^2 - 4)}{5 - 10}$

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Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples
ALGEBRAIC EXPRESSION	
SUBSTITUTION *Property*	If _____, then _____
EVALUATING *Expressions*	**To evaluate an expression with variable replacements:** • _____ the variables with their given values. • Each time you substitute a variable with a number, put _____ around the number! • Follow the _____ _____ _____ to evaluate!
EXAMPLES	**Directions:** Evaluate each expression using the variable replacements. **1.** $ab^2 + c$ if $a = 2$, $b = 4$, $c = 7$ **2.** $3x^2 - 4x$ if $x = -2$
YOU TRY!	**Directions:** Evaluate each expression using the variable replacements. **3.** $a^2b - b^2$ if $a = 3$ and $b = -4$ **4.** $a^2b - b^2$ if $a = 4$ and $b = -7$ **5.** $-y^2 - 3xy$ if $x = -4$ and $y = 2$ **6.** $-y^2 - 3xy$ if $x = -\frac{5}{6}$ and $y = -12$

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7. $\frac{a^2+b^2}{a-b}$ if a = 3 and b = 4

8. $\frac{a^2+b^2}{a-b}$ if a = -3 and b = -5

9. $x^2 - 2(x-y) - z^3$ if x = 2, y = -2, and z = -3

10. $x^2 - 2(x-y) - z^3$ if x = -3, y = 4 and z = -1

11. $-a^2 - 2bc - |c|$ if a = -2, b = 3, and c = -3

12. $-a^2 - 2bc - |c|$ if a = -3, b = -5, and c = 2

Summary: _____

_____

_____

_____

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Name: _____ **Unit 1:** Algebra Basics

Date: _____ **Bell:** _____ **Homework 5:** Evaluating Expressions

Directions: Evaluate each expression for the given variable values.

1. $n^2 - 3n + 8$ if $n = 4$

2. $\frac{x^2 - 4y}{2}$ if $x = 4$ and $y = -3$

3. $4|m - n|$ if $m = -7$ and $n = 2$

4. $19 - x^2$ if $x = -5$

5. $s^2t - 10$ if $s = -8$ and $t = \frac{3}{4}$

6. $4p^2 + 7q^3$ if $p = -3$ and $q = -2$

7. $\frac{(a+b)^5}{a-2b^2}$ if $a = 6$ and $b = -8$

8. $\frac{5c^2 - d^2 + 3}{2c - 4d}$ if $c = -1$ and $d = -4$

9. $-q^2 - r^2 + 3s$ if $q = 9$, $r = -6$, $s = -20$

10. $3|x + y|^2 - (xy)^2$ if $x = 3$ and $y = -5$

11. $2x^2 - 5xy - y^3$ if $x = -3$ and $y = -2$

12. $-a^2 + 7b^4 - 2c^3$ if $a = -4$, $b = -2$, $c = -3$

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Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples
MATRICES	
ELEMENTS	
ROWS	
COLUMNS	
DIMENSIONS	
EXAMPLE	**Sue and Eva made a table of their science quiz grades:**
IDENTIFYING ELEMENTS	Element _____ is used to identify a specific element where a is the _____ number and b is the _____ number.
EXAMPLES	**Directions:** Given matrix M to the left, identify each element.

Sue	B	C	A	B	A	B	A	C	B	A	B	C
Eva	A	B	B	A	A	C	B	B	A	A	B	B

1. Create a matrix to organize the data:

2. Give the dimensions: _____

$$M = \begin{bmatrix} 5 & 3 & 1 & 0 \\ -2 & -8 & 2 & 11 \\ 7 & 9 & 1 & -5 \end{bmatrix}$$

3. Element $X_{(1, 2)}$ = _____

4. Element $X_{(3, 2)}$ = _____

5. Element $X_{(2, 4)}$ = _____

6. Element $X_{(1, 3)}$ = _____

7. Element _____ = -5

8. Element _____ = -2

9. Element _____ = 7

10. Element _____ = -8

11. What are the dimensions of matrix M? _____

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ADDITION & SUBTRACTION

- Matrices must have the same _____ in order to add or subtract them.
- Simply add or subtract _____ _____ to create the new matrix.

EXAMPLES

Using the matrices below, perform each operation, if possible.

$$A=\begin{bmatrix}7 & 19\\ 4 & -6\\ -1 & -2\end{bmatrix} \qquad B=\begin{bmatrix}-12 & 9\\ -5 & -1\\ 16 & 2\end{bmatrix} \qquad C=\begin{bmatrix}1 & 13 & -4\\ 2 & -8 & 7\end{bmatrix} \qquad D=\begin{bmatrix}-14 & 0\\ 1 & -5\\ 10 & 9\end{bmatrix}$$

12. $B+D$

13. $B-A$

14. $C+B$

15. $A-D$

SCALAR MULTIPLICATION

A matrix can be multiplied by a _____ (any real number) by _____ the scalar to each separate element.

EXAMPLES

Using the matrices given above, perform each operation.

16. $4C$

17. $-2A$

18. $3D+B$

19. $5D-2B$

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Name: _____ **Unit 1:** Algebra Basics

Date: _____ **Bell:** _____ **Homework 6:** Matrices

Directions: State the dimensions of each matrix.

1. $\begin{bmatrix} 3 & -4 & -9 \\ 2 & -7 & 0 \end{bmatrix}$

2. $\begin{bmatrix} 20 \\ -6 \end{bmatrix}$

3. $\begin{bmatrix} 10 & 7 \\ 0 & 16 \\ -3 & -8 \\ -5 & 1 \end{bmatrix}$

4. $\begin{bmatrix} 10 & -4 & -20 & 15 \end{bmatrix}$

Directions: Add/Subtract the following matrices (if possible)

5. $\begin{bmatrix} -2 & 6 \\ 1 & 4 \end{bmatrix} + \begin{bmatrix} -3 & 11 \\ 2 & 7 \end{bmatrix}$

6. $\begin{bmatrix} 14 & -3 & 1 \end{bmatrix} - \begin{bmatrix} 2 & -4 & -7 \end{bmatrix}$

7. $\begin{bmatrix} 2 & -5 \\ 6 & 8 \end{bmatrix} + \begin{bmatrix} 10 \\ 15 \end{bmatrix}$

8. $\begin{bmatrix} 13 & -10 & 12 \\ 6 & 3 & 30 \\ 0 & -5 & 5 \end{bmatrix} - \begin{bmatrix} 1 & 6 & -7 \\ 2 & -5 & 8 \\ -3 & 4 & 9 \end{bmatrix}$

Directions: Use scalar multiplication to evaluate the following matrices.

9. $-3\begin{bmatrix} 12 & -5 \end{bmatrix}$

10. $8\begin{bmatrix} -1 & 2 \\ 0 & 4 \end{bmatrix}$

Directions: If A = $\begin{bmatrix} 7 & 3 \\ -4 & -1 \end{bmatrix}$, B = $\begin{bmatrix} 9 & 6 \\ 3 & 10 \end{bmatrix}$, and C = $\begin{bmatrix} 0 & -11 \\ 2 & 8 \end{bmatrix}$, find the following:

11. 4A – 2B

12. 2C – A

13. 3A + B – C

14. B − A

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Name: _____ **Algebra I**

Date: _____ **Bell:** _____ **Unit 1:** Algebra Basics

Quiz 1-2: Order of Operations, Matrices, & Evaluating Expressions

Part I: Absolute Value & Order of Operations Show all work for those with several steps!

1. |6| + |-2|

2. $2^3 - 5 \cdot (6 + 4) \div 2$

3. $86 - \left[(4 - 9)^2 \cdot 3\right]$

4. |2 + 3(-4)|

5. $\dfrac{36 - 3 \cdot 4}{15 - 9 \div 3}$

6. $\dfrac{1 + 3 \cdot 5}{18 - 4^2} - \left|-11\right|$

Answers

1. _____

2. _____

3. _____

4. _____

5. _____

6. _____

Part II: Matrices If A = $\begin{bmatrix} 7 & -2 & 3 \\ -4 & 11 & 0 \end{bmatrix}$, B = $\begin{bmatrix} 15 \\ -6 \end{bmatrix}$, C = $\begin{bmatrix} -9 \\ 24 \end{bmatrix}$ and D = $\begin{bmatrix} -6 & 12 & 8 \\ 0 & -1 & 17 \end{bmatrix}$ find:

7. $A + D$

8. $C - B$

9. $B + D$

10. $-3B$

11. $2A - D$

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Part III: Evaluating Expressions If w **= -1**, x **= 6**, and y **= -3**, find the following:

12. $2y - 5w$

13. $-\frac{7}{3}x + y^2$

14. $3w^2 - 5w + 1$

15. $2|y| - x^2$

16. $-x^2 - 7xy + w^3$

Answers
12. _____
13. _____
14. _____
15. _____
16. _____
B. _____

B. $7\left[4^2 - 6\left|5 + (2^3 - 18) + 16 \div 8\right|\right]$ ***Show all steps to receive credit!***



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Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples

Parts of an Expression

$7x + 3 - 5x + 8 + 4x - 1$

- **Variable Terms** (Terms WITH a variable)
  - **Coefficients** (Number NEXT TO a variable)
- **Constant Terms** (Terms WITHOUT a variable)

Combining Like Terms

You can simplify an algebraic expression by **combining like terms.** This means to combine common variable terms and constant terms.

Example: Simplify the expression below:

$7x + 3 - 5x + 8 + 4x - 1 =$ _____

Examples

Directions: Simplify each expression.

1. $3x + 6x$	**2.** $9a - 10a$
3. $7m - 5m - 6$	**4.** $9 - 6w + 5$
5. $7p - 1 - 9p + 5$	**6.** $5h - 6 - 8 + 7h$
7. $5x + 7y - x + 2y + 3x$	**8.** $26a + 2b - 4a + 10b$

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with Exponents

Terms with the same exponents or combination of variables and exponents are considered like terms. When combined, be sure to NOT change their exponents!

Examples

9. $5x^2 + 7x^2$

10. $-15n^5 + 4n^5$

11. $8c^3 - 7c^3 + 16c^3$

12. $8k^2 - k - 5k + 7 - 2k^2$

13. $-4a^2b + 7 + 10a^2b$

14. $x^2 + 2xy - 7xy + 4y^2$

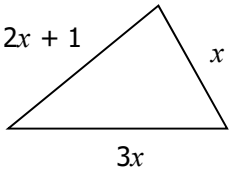
15. $a^2 + 7ab - 10ab + b^2$

16. $4x^2 - 7 + 2x - 3x^2 + 8x - 6$

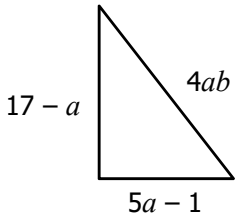
Geometric Applications

Directions: Give the perimeter of each figure as a simplified expression.

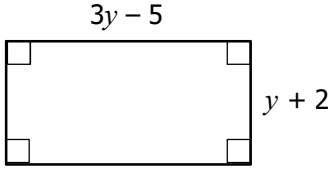
17.



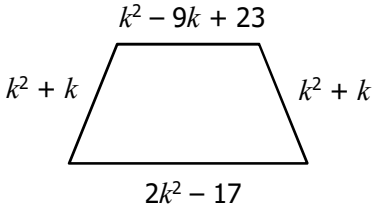
18.



19.



20.



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Name: _____ **Unit 1:** Algebra Basics

Date: _____ **Bell:** _____ **Homework 7:** Combining Like Terms

Directions: Simplify each expression by combining like terms.

1. $8y - 9y$

2. $3x - 7x + 1$

3. $8a - 6 + a - 1$

4. $-2y + 14 - 9y + 5$

5. $-x - 2 + 15x$

6. $-7w - 5 - 7w + 9$

7. $8d - 4 - d - 2$

8. $40 - 9x + 4x + 5$

9. $9a + 8 - 2a - 3 - 5a$

10. $4xy - 6xy + 5y - 9x$

11. $7x + 2y + 5 - 5x + 9y - 1$

12. $-5f - 7g + f - 5g - 4 + f$

13. $-6 + 3a - b - 4a + 4b + 5$

14. $6n^2 - 2n - 3 + 5n^2 - 9n - 6$

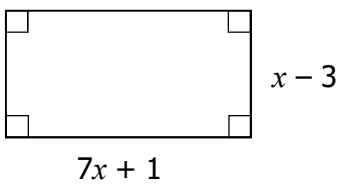
15. $8c^2 - c + 3 + 2c^2 - c + 2$

16. $-12x^2y - 3x^2y$

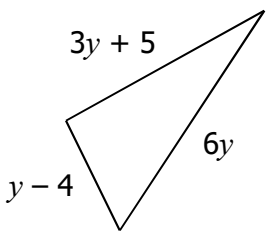
17. $6a^2b^2 + 3ab^2 - a^2b^2 - 4ab^2$

18. $19x^2y - 4xy^2 + 5xy^2 - 16x^2y$

19. Write the perimeter of the rectangle as a simplified expression.



20. Write the perimeter of the triangle as a simplified expression.



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Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples		
Distributive Property **Recall:** $a(b+c) =$ or $a(b-c) =$	**1.** $7(x + 4)$	**2.** $2(b - 3)$	**3.** $-4(y + 3)$
	4. $-5(m - 2)$	**5.** $-(y - 9)$	**6.** $8(4 - b)$
	7. $-4(-w - 10)$	**8.** $5(2m - 3)$	**9.** $-3(2x - 4)$
	10. $3(m + n)$	**11.** $x(y + 4)$	**12.** $d(c - 4)$
	13. $6a(b + 2c)$	**14.** $2(x - y + 8)$	**15.** $x(2y - 5z + 1)$
	16. $\frac{1}{4}(8x - 4)$	**17.** $\frac{1}{6}\left(\frac{3}{5}x + 18\right)$	**18.** $-\frac{1}{8}\left(\frac{4}{5}x - 24\right)$
Distribute AND Combine!	To **simplify an expression** means to ensure there are **no parentheses** and **no like terms**. In order to do this, distribute first (if needed),then combine like terms.		
Examples	**19.** $8(2x - 3) - 6x$	**20.** $9(2k - 4) - 2(7k - 12)$	
	21. $10 - (y - 6) - y$	**22.** $6 + 8(4w - 7) - (2w + 1)$	
	23. $\frac{9}{4}\left(\frac{10}{3}x - 4\right) - \frac{1}{4}(2x - 8)$	**24.** $\frac{7}{8}\left(\frac{4}{5}x + 16\right) + \frac{3}{5}\left(\frac{1}{2}x - 10\right)$	

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Name: _____ **Unit 1:** Algebra Basics

Date: _____ **Bell:** _____ **Homework 8:** Simplifying Expressions

Directions: Simplify each expression by distributing.	
1. $8(x + 5)$ = _____	**2.** $3(2y - 7)$ = _____
3. $-2(3m + 9)$ = _____	**4.** $-6(z - 1)$ = _____
5. $-(5x + 1)$ = _____	**6.** $-(x + 4)$ = _____
7. $b(a - 6)$ = _____	**8.** $7a(3b - 2c + 4)$ = _____
9. $\frac{1}{2}(8x+2)$ = _____	**10.** $\frac{2}{3}\left(\frac{1}{4}x-6\right)$ = _____

Directions: Simplify each expression by distributing and combining like terms.	
11. $-7(w - 4) + 3w - 27$	**12.** $6(x + 1) - 5(x + 2)$
13. $-5(k + 6) + 7(k - 4)$	**14.** $8(x + 5) - 4(x + 4)$
15. $-9(m + 2) + 4(6 - 7m)$	**16.** $-3(-9v - 4) - 2(v - 2)$
17. $-4(8 - c) - (3c + 7)$	**18.** $6(p + 6) - (p - 6)$
19. $21 - 8(w + 3) + 3 + 7w$	**20.** $\frac{1}{2}\left(\frac{3}{5}x-8\right)+\frac{7}{4}\left(\frac{2}{5}x+4\right)$

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Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples	
Key Words & Phrases	**ADDITION** + • _____ • _____ • _____ • _____ • _____	**SUBTRACTION** − • _____ • _____ • _____ • _____ • _____
	MULTIPLICATION ● • _____ • _____ • _____ • _____ • _____	**DIVISION** / • _____ • _____ • _____ • _____ • _____
	(Parentheses Phrases) • _____ • _____	**"Turn Around" Phrases** • _____ • _____
Examples	**Directions:** Translate each expression or equation.	
	1. "the product of a number and 3"	**2.** "the difference between a number and 10"
	3. "the quotient of a number and -5"	**4.** "four subtracted from a number"
	5. "the sum of twice a number and 7"	**6.** "the product of a number and 10, increased by 1"

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	7. "four less than a number squared"		**8.** "twice a number divided by 5"	
Equations	**9.** "the sum of twice a number and 10 is 36"		**10.** "three times the difference of a number and 7 is 18"	
	11. "four times a number divided by 8 is 3"		**12.** "five times the sum of a number and 6 is 48"	
	13. "one half of a number is 8 less than the number itself"		**14.** "the product of a number and 3 is the same as the sum of that number and six"	
	15. "fifteen decreased by the quotient of a number and 3 is 20"		**16.** "four less than the product of a number and 7 is eight more than that number"	
Inequalities	**LESS THAN**	**LESS THAN OR EQUAL TO**	**GREATER THAN**	**GREATER THAN OR EQUAL TO**
Examples	**17.** "a number is less than -10"		**18.** "a number is less than or equal to 8"	
	19. "a number is greater than or equal to -4"		**20.** "a number is at most 45"	
	21. "a speed limit of 55 mph"		**22.** "a number is at least 16"	
	23. "a number is no less than 10"		**24.** "a number is no more than 26"	
	25. "a maximum number of 25"		**26.** "a minimum number of 5"	

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Name: _____ **Unit 1:** Algebra Basics

Date: _____ **Bell:** _____ **Homework 9:** Translating Expressions Equations, & Inequalities

Directions: Translate each expression.	
1. "the sum of a number and 4."	**2.** "the quotient of a number and -2."
3. "the product of a number and 5."	**4.** "the difference of a number and 7."
5. "a number squared increased by 1"	**6.** "three less than twice a number"

Directions: Translate each equation. ***DO NOT SOLVE!***	
7. "twice a number divided by 6 is 42."	**8.** "two times the sum of number and 5 is 20"
9. "four times the difference of a number and 7 is 32"	**10.** "one-third of a number is 12 less than the number itself"
11. "ten increased by the quotient of a number and 2 is -15"	**12.** "one less than the product of a number and 3 is 5 more than the number itself"

Directions: Translate each inequality.	
13. "a number less than 15"	**14.** "a number greater than or equal to -3"
15. "a number that is at least 90"	**16.** "a speed limit of 65 mph"
17. "a number that is no more than 20"	**18.** "a maximum number of 50"
19. "a minimum number of 10"	**20.** "a number no less than 25"

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WARM-UP!!!

Translate the following expressions, equations, & inequalities:

1. "The product of a number and 7" _____

2. "Nine subtracted from twice a number" _____

3. "One less than the quotient of a number and -5. _____

4. "Three times the sum of a number and 10 is 24" _____

5. "The difference of twice a number and 3 is -21." _____

6. "One-third of a number increased by 8 is 37." _____

7. "x is at most 6" _____

8. "You must be at least 18 years old to vote" _____

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Name: _____ **Algebra I**

Date: _____ **Bell:** _____ **Unit 1:** Algebra Basics

Quiz 1-3: Simplifying & Translating Expressions

Part I: Simplifying Expressions Simplifying by distributing and/or combining like terms.

1. $\frac{-2}{3}\left(\frac{5}{6}x-3\right)$

2. $3c(5a + 2b - 1)$

3. $5x - 4 - 7x + 14$

4. $-3 - 5x - 5y + 8x + 7y + 8$

5. $4x^2y + 2xy - 6 + 5x^2y - 8xy + 13$

6. $2(x - 1) + 5(x + 9)$

7. $3(x + 9) - (x - 6)$

8. $3(5x + 9 - 4x)$

9. $a + 3a - 4(9 - a)$

Answers

1. _____

2. _____

3. _____

4. _____

5. _____

6. _____

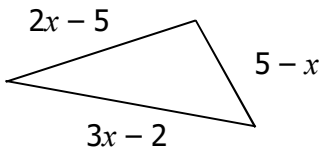
7. _____

8. _____

9. _____

10. _____

10. Write the perimeter of the triangle as a simplified expression.



Part II: Translating Expressions, Equations, and Inequalities.

EXPRESSIONS:

11. "the product of a number and 4"

12. "five times the difference of a number and 3"

13. "seven less than a number squared"

Answers

11. _____

12. _____

13. _____

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EQUATIONS:

14. "the quotient of a number and 2, increased by 9 is -15"

15. "one-fourth of a number minus 8 is -23"

16. "the sum of twice a number and 6 is the same as 3 subtracted from the number itself"

INEQUALITIES

17. "a number greater than -5"

18. "a number that is at most 30"

19. "a minimum number of 17"

20. "a number that is at least 90"

Answers

14. _____

15. _____

16. _____

17. _____

18. _____

19. _____

20. _____

Part III: REVIEW! Circle the best answer.

21. Which number is an integer?

A. -18 **B.** $\frac{4}{7}$ **C.** $\sqrt{15}$ **D.** 0.35

22. Which set of real numbers does **NOT** contain the number 0?

A. Whole Numbers
B. Natural Numbers
C. Integers
D. Rational Numbers

23. Which property justifies the statement $(a - b) \cdot 0 = 0$?

A. Inverse Property
B. Associative Property
C. Distributive Property
D. Zero Product Property

24. Which property justifies the statement if y = 7, then 7 = y?

A. Reflexive Property
B. Identity Property
C. Symmetric Property
D. Transitive Property

25. Which set of numbers is closed under subtraction?

A. Natural Numbers
B. Integers
C. Irrational Numbers
D. Odd Numbers

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Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples	
One-Step Equations	**1.** $m + 12 = 10$	**2.** $-2 = g - 9$
	3. $-7y = -91$	**4.** $\frac{a}{9} = -4$
Fractions To "get rid" of a fraction, multiply by the _____!	**5.** $\frac{2}{3}x = 10$	**6.** $\frac{4}{9}w = -8$
	7. $-\frac{6}{5}k = 12$	**8.** $-\frac{1}{2}m = -9$
Two-Step Equations	**To Solve a Two-Step Equation:** **1.** Undo the **Addition/Subtraction** (to remove constant term) **2.** Undo the **Multiplication/Division** (to remove coefficient)	
	9. $6x + 8 = 50$	**10.** $2n - 5 = 11$
	11. $13 = -4k + 9$	**12.** $7 - 3y = 34$

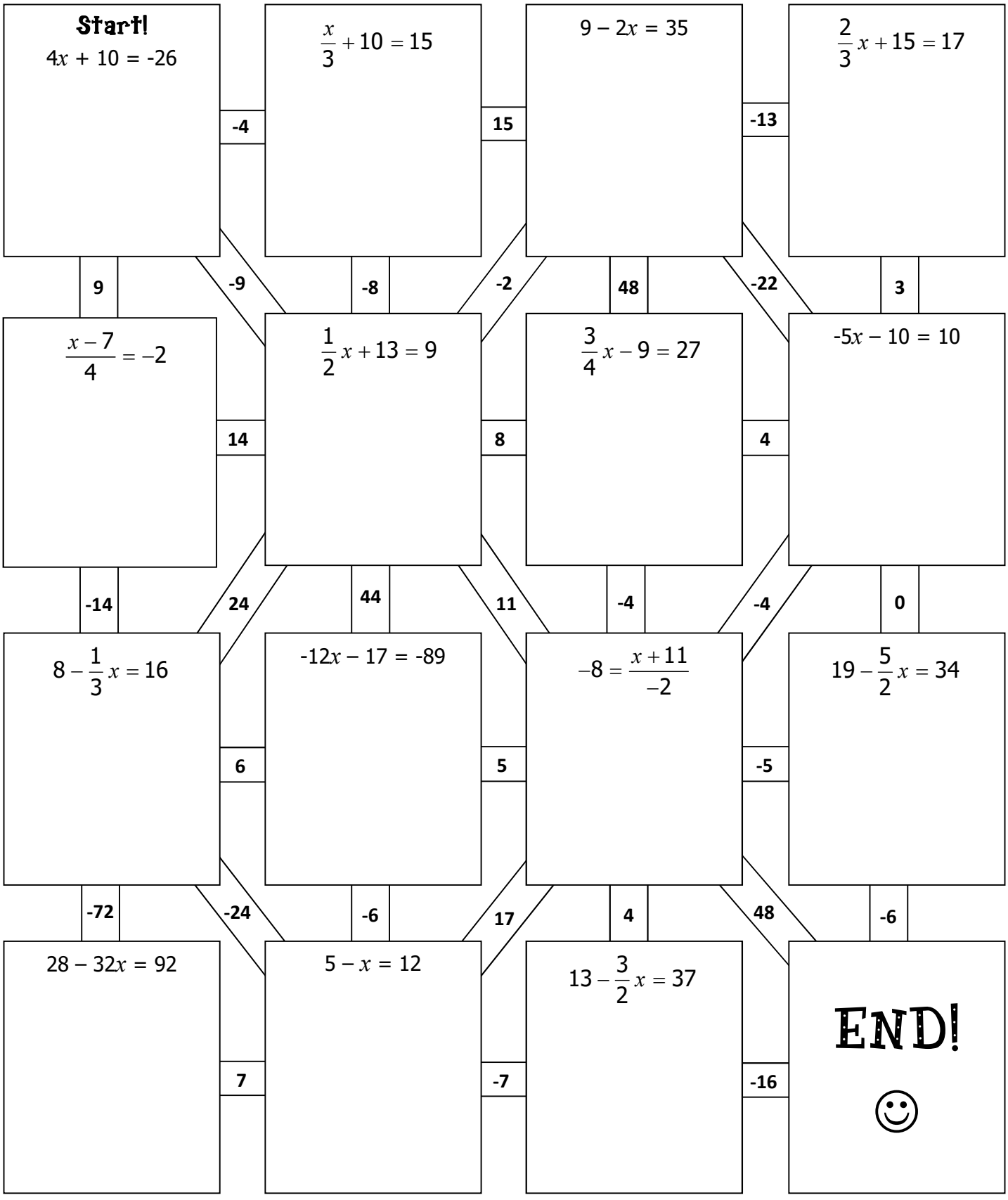
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	13. $\frac{x}{2} - 7 = 9$	**14.** $11 = \frac{c}{-5} + 8$
	15. $\frac{3}{5}x + 22 = 28$	**16.** $-\frac{1}{3}m + 1 = -7$
	17. $-10 + \frac{7}{4}p = -38$	**18.** $15 = 9 - \frac{1}{2}x$
Watch Out!	**The examples below are different in that the multiplication/division is done FIRST, followed by the addition/subtraction.**	
	19. $\frac{x + 11}{8} = -3$	**20.** $\frac{n - 5}{-2} = -7$
	21. $1 = \frac{a - 13}{-6}$	**22.** $4 = \frac{w + 8}{9}$

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Two-Step Equation Maze!

Directions: Use your solutions to navigate through the puzzle. **SHOW ALL STEPS!!!!**



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Name: _____ **Unit 1:** Algebra Basics

Date: _____ **Bell:** _____ **Homework 10:** One & Two-Step Equations

Directions: Solve each equation. **SHOW ALL STEPS!**

1. $y - 17 = -37$	**2.** $-12 + k = 18$	**3.** $5m = -20$
4. $\frac{x}{9} = 3$	**5.** $\frac{4}{5}z = 16$	**6.** $3p - 4 = 11$
7. $-5a + 7 = -38$	**8.** $31 - 12n = 211$	**9.** $\frac{x}{3} - 15 = -7$
10. $\frac{f}{-4} + 11 = 5$	**11.** $\frac{4}{3}v + 25 = 33$	**12.** $\frac{5}{8}c + 13 = 18$
13. $-\frac{2}{3}w + 53 = 59$	**14.** $11 = \frac{4}{5}h - 1$	**15.** $-\frac{3}{8}r + 41 = -22$
16. $34 = 28 - \frac{2}{5}y$	**17.** $\frac{n+4}{-2} = -9$	**18.** $-5 = \frac{x-1}{8}$

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Name:	Date:
Topic:	Class:

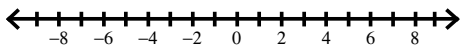
Main Ideas/Questions	Notes/Examples

Rules for GRAPHING INEQUALITIES

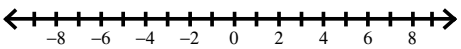
- ✓ Use an _____ _____ for _____ or _____ signs.
- ✓ Use a _____ _____ for _____ or _____ signs.

Graph the following inequalities on the number line.

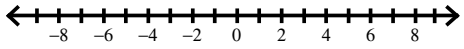
1. $x < 3$



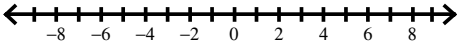
2. $x > -1$



3. $x \geq 7$

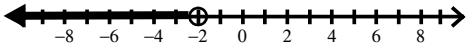


4. $x \leq -5$

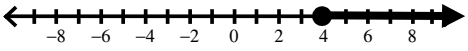


Write an inequality given the graph.

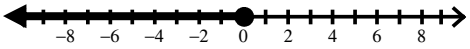
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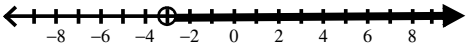
6.



7.



8.



Rules for SOLVING INEQUALITIES

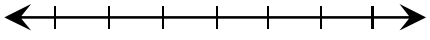
You must _____ the inequality symbol when you

_____ or _____

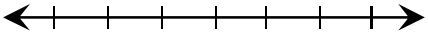
by a _____ _____!

Solve and graph the following inequalities.

9. $3x + 2 < -4$

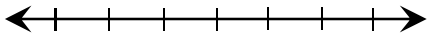


10. $\frac{x}{6} + 2 > -1$

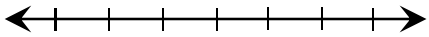


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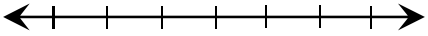
11. $-7x + 8 \leq 1$



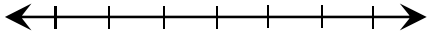
12. $\frac{x}{-8} - 5 < -2$



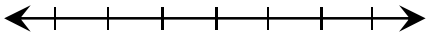
13. $-2x - 5 \leq 19$



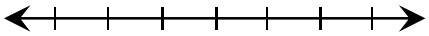
14. $3 - 4x < 31$



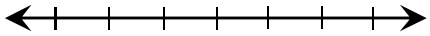
15. $\frac{3}{4}x + 5 > 11$



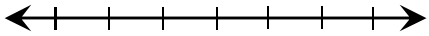
16. $-\frac{5}{6}x - 1 \geq 14$



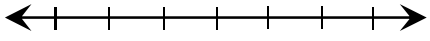
17. $5x - 2 \leq -7$



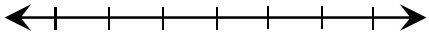
18. $6 - x \leq 14$



19. $4 - \frac{1}{3}x > 11$



20. $\frac{x+11}{-2} < -2$



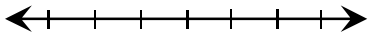
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Name: _____ **Unit 1:** Algebra Basics

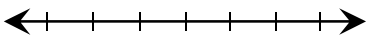
Date: _____ **Bell:** _____ **Homework 11:** Solving & Graphing Inequalities

Directions: Solve and graph the following inequalities. **SHOW ALL STEPS!**

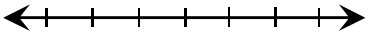
1. $3x + 1 < 10$



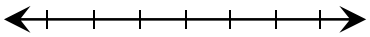
2. $\frac{x}{3} + 10 \geq 15$



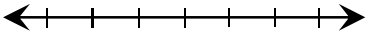
3. $-2x + 22 < 18$



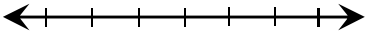
4. $7x + 11 > -31$



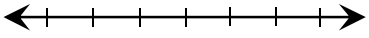
5. $\frac{x}{5} - 3 \leq 6$



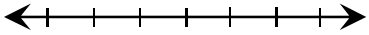
6. $18 - 2x > 4$



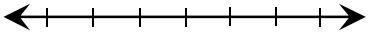
7. $\frac{x}{-2} + 1 > 4$



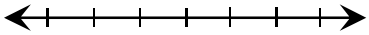
8. $4 - \frac{x}{7} \leq 5$



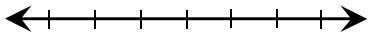
9. $-3x - 4 < 8$



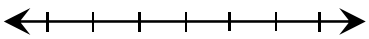
10. $\frac{5}{8}x - 1 \geq 9$



11. $-\frac{3}{4}x + 2 > 11$



12. $\frac{x+15}{-3} > -7$



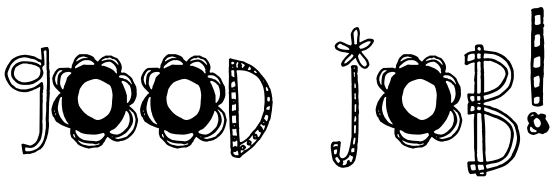
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Translate... and Solve!

1 "Twice a number plus 4 is 8"		2 "The product of 4 and a number plus 17 is 5."	
3 "Twice a number minus 5 is 7"		4 "The quotient of a number and 6 minus 3 is 1"	
5 "Two more than 4 times a number is -18"		6 "Seven less than twice a number is 5"	
7 "Five less than the quotient of a number and 3 is -7"		8 "One subtracted from the product of 4 and a number is 11"	
9 "The quotient of a number and -9, increased by 10 is 11"		10 "Half of a number decreased by 8 is -3"	

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11 "The sum of three-fourths of a number and 1 is -5"		12 "Ten minus one-third of a number is 4"	
13 "Four more than the quotient of a number and 3 is at least 9"		14 "Negative three times a number increased by seven is less than -11"	
15 "Seven more than one-sixth of a number is greater than three"		16 "The sum of 5 and the twice a number is at most 27"	
17 "Four less than two-fifths of a number is no less than 2"		18 "Three-fourths of a number decreased by 10 is greater than or equal 5"	
19 "The quotient of a number and -5, increased by 7 is at minimum 1"		20 "One less than half a number is a maximum of 19"	



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Unit 1 Test Study Guide
Algebra Basics

Name: _____

Date: _____ Block: _____

Topic 1: The Real Number System

List ALL sets to which each number belongs. (Use *R*, *I*, *Q*, *Z*, *W*, *N*)		
1. $-\frac{14}{2}$	**2.** $\sqrt{64}$	**3.** 0
4. π	**5.** $0.\overline{45}$	**6.** $\frac{3}{8}$

Place the LETTER of each value in its location in the real number system below.

A. $-0.\overline{2}$ **B.** 18

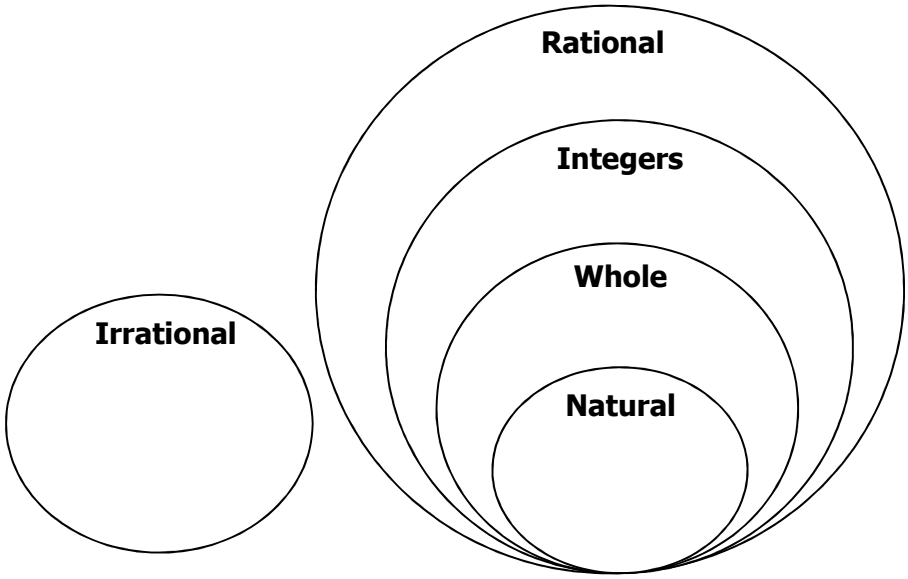
C. $-\sqrt{100}$ **D.** π

E. 0 **F.** $2\frac{1}{6}$

G. -5 **H.** 4.03

I. $-\sqrt{72}$ **J.** $\sqrt{\frac{4}{9}}$

K. $\frac{36}{9}$



Topic 2: Properties

Identify the property shown below.	
7. $4 + (x + y) = (4 + x) + y$	**8.** $\frac{2}{5} \cdot \frac{5}{2} = 1$
9. if $\sqrt{49} = 7$ and $7 = 3 + 4$, then $\sqrt{49} = 3 + 4$	**10.** -28 = -28
11. $8x^2 \cdot 1 = 8x^2$	**12.** $10y + (-10y) = 0$
13. $(a + 4) \cdot 0 = 0$	**14.** $-5(x + 7) = -5x - 35$
15. $(x + 2) + y = (2 + x) + y$	**16.** if $x = -1$, then $-1 = x$

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Answer true or false. Provide an example if false.
17. The set of negative numbers are closed under division.
18. The set of whole numbers are closed under subtraction.
19. The set of integers are closed under multiplication.

Topic 3: Absolute Value & Order of Operations

20. $\lvert -21 \rvert + \lvert 5 \rvert$	**21.** $\lvert -10 - 4 \rvert$
22. $10 - 2 \cdot 3^2 + 13$	**23.** $[(-5+1) \div 2]^3 - \lvert -7 \rvert$
24. $\dfrac{25 + 5^2 \div 5}{7 - 12 \div 4 + 2}$	**25.** $\dfrac{(3-7)^2 + 11}{\lvert -2 \rvert + \lvert -1 \rvert}$

Topic 4: Evaluating Expressions

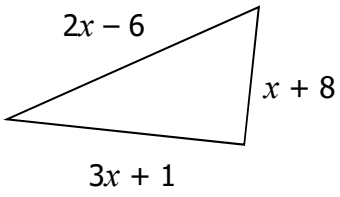
26. $3x^3 - 8y^2$ if $x = 2$ and $y = -3$	**27.** $\lvert a - 4b \rvert$ if $a = 7$ and $b = 2$
28. $w^2 + 2xy$ if $x = -3$, $w = -2$, and $y = 1$	**29.** $\dfrac{7c^2 + 5}{4a - b}$ if $a = 1$, $b = -5$ and $c = -4$

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Topic 5: Translating Expressions, Equations, & Inequalities

30. "The product of a number and 7, increased by three."	**31.** "One less than twice a number."
32. "Four times the difference of a number and nine is -30."	**33.** "Five more than the quotient of a number and eight is 42."
34. "The auditorium can hold a maximum of 150 people."	**35.** "You must be at least 25 to rent a car."

Topic 6: Simplifying Expressions

36. $n + 4 - 9 - 5n$	**37.** $7x - 4y - 8 - 3x - 8y + 12$
38. $18 - 2(4x + 7) + 5x$	**39.** $9(x - 3) - (x + 2)$
40. Write the perimeter of the triangle in simplest form.  	**41.** Write the perimeter of the rectangle in simplest form. 

Topic 7: Two-Step Equations

42. $3x - 7 = 11$	**43.** $\frac{x}{4} + 8 = 3$

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44. $\frac{2}{5}x - 3 = -17$

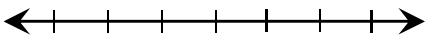
45. $9 - 7x = 16$

46. $9 - \frac{3}{4}x = 57$

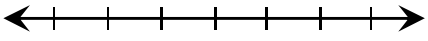
47. $\frac{x-17}{3} = -3$

Topic 8: Solving & Graphing Inequalities

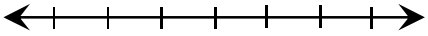
48. $3x + 5 < 14$



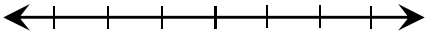
49. $-5x + 2 \geq 52$



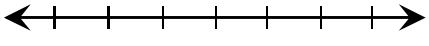
50. $\frac{x}{-2} + 11 \leq 15$



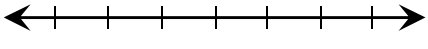
51. $4x - 1 \leq -37$



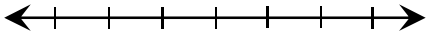
52. $\frac{5}{8}x - 1 \leq 9$



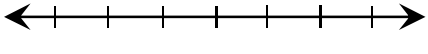
53. $4 - \frac{x}{7} < 5$



54. $18 - 2x > 4$



55. $8 - \frac{3}{2}x \geq -13$



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Name: _____

Date: _____ **Bell:** _____

Algebra I Unit 1 Test

(Algebra Basics)

SHOW ALL WORK NEEDED TO ANSWER EACH QUESTION!

PLACE YOUR FINAL ANSWER IN THE BOX. GOOD LUCK! ☺

1. Which is the smallest set of real numbers that contains the value below?

$$-\frac{18}{6}$$

A. Irrational Numbers
B. Rational Numbers
C. Natural Numbers
D. Integers

2. Which set contains the value below?

$$\sqrt{50}$$

A. Irrational Numbers
B. Natural Numbers
C. Rational Numbers
D. Integers

3. The set below only contains which types of numbers?

$$\{-1, 5, \frac{1}{2}, 15, 3.75, 36, \sqrt{81}, 100\}$$

A. Irrational Numbers
B. Rational Numbers
C. Integers
D. Natural Numbers

4. Select all sets to which the value below belongs.

$$\sqrt{2} - \sqrt{2}$$

☐ Real Numbers
☐ Irrational Numbers
☐ Rational Numbers
☐ Integers
☐ Whole Numbers
☐ Natural Numbers

5. Which of the following is ***true*** regarding number sets?

A. All integers are whole numbers.
B. All irrational numbers are real numbers.
C. All real numbers are integers.
D. All rational numbers are natural numbers.

6. Which property justifies the statement below?

$$x(y - 3) = xy - 3x$$

A. Associative Property
B. Transitive Property
C. Distributive Property
D. Commutative Property

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7. Which property justifies the statement below?

$$(a + b) + 0 = (a + b)$$

A. Reflexive Property
B. Identity Property
C. Symmetric Property
D. Associative Property

8. Which property justifies the statement below?

if $9x = 27$, then $27 = 9x$

A. Commutative property
B. Identity Property
C. Symmetric Property
D. Transitive Property

9. Give an example of the reflexive property.

10. Give an example of the associative property.

11. Integers are ***not*** closed under which operation?

A. Addition
B. Subtraction
C. Multiplication
D. Division

12. Which of the following is a ***false*** statement?

A. Natural numbers are closed under division.
B. Integers are closed under subtraction.
C. Even numbers are closed under multiplication.
D. Real numbers are closed under addition.

13. Simplify the expression below.

$$4 + 2^3 - |-4|$$

14. Simplify the expression below.

$$3^2 - 3 \cdot [(5^2 - 9) \div 2]$$

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15. Simplify the expression below

$$\frac{(3-13)^2+14}{4^2-5\cdot 2}$$

16. Evaluate the expression below if a = -8, b = 17, and c = 21

$$a^2-(b+c)$$

17. Evaluate the expression below if x = -1 and y = 3

$$3x^2-y^2$$

18. Evaluate the expression below if a = 8 and b = -5

$$|a|-|b|$$

19. Translate statement below.

"One less than twice a number."

A. $1-n^2$

B. n^2-1

C. $1-2n$

D. $2n-1$

20. Translate the statement below.

"Five times the difference of a number and 3 is 17."

A. $5n-3=17$

B. $5(n-3)=17$

C. $\frac{5n}{3}=17$

D. $5\left(\frac{n}{3}\right)=17$

21. Translate the statement below.

"A number is no more than 50"

A. $x<50$

B. $x\leq 50$

C. $x>50$

D. $x\geq 50$

22. Translate the statement below

"Your grade must be at least 64 to pass this class"

A. $g<64$

B. $g\leq 64$

C. $g>64$

D. $g\geq 64$

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23. Simplify the expression below.

$$2m - 16 + 5m + 45$$

A. $7m + 29$

B. $7m + 61$

C. $-3m + 29$

D. $-3m + 61$

24. Simplify the expression below.

$$2x - 4y + 6 + 3x - 9y - 4$$

A. $5x + 13y + 2$

B. $5x - 13y + 2$

C. $5x - 5y + 2$

D. $-8xy + 2$

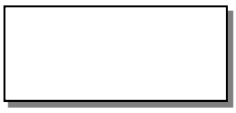
25. Simplify the expression below completely.

$$10 - 4(2x + 7)$$

26. Simplify the expression below completely.

$$4(5w - 3) - (w - 1)$$

27. Give the perimeter of the rectangle below in simplest form.



28. Identify the first step to solve the equation below.

$$\frac{x}{-4} + 7 = -1$$

A. Add 4

B. Subtract 7

C. Multiply by -4

D. Add 1

29. Solve the equation below.

$$5x + 1 = -49$$

30. Solve the equation below.

$$\frac{2}{3}x - 5 = 7$$

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31. Solve the equation below.

$$2 - 4a = 18$$

32. Solve the equation below.

$$12 - \frac{1}{9}n = 17$$

33. Solve the inequality below.

$$4x - 1 \leq -29$$

34. Solve the inequality below.

$$\frac{x - 13}{-6} < 3$$

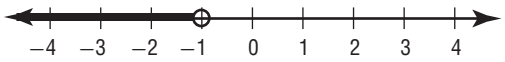
35. Which graph represents the solution to the inequality below?

$$4 - 2x > -2$$

A.



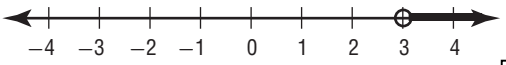
B.



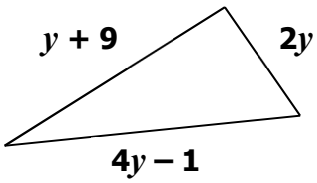
C.



D.



B. Rob would like to create a triangle that is seven less than twice the perimeter of the triangle below. Give an expression for the perimeter of the new triangle in simplest form.



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The Real Number System

REAL NUMBERS

R = all numbers

Decimals that never end and have no repeating pattern. *Think ***MUMBO JUMBO NUMBERS***

IRRATIONAL NUMBERS

I = $\pi, \sqrt{2}, -\sqrt{101}$

All fractions and decimals that end or have a repeated pattern.

RATIONAL NUMBERS

Q = $\frac{1}{3}, -0.67, \sqrt{16}$

INTEGERS

Z = $\ldots, -2, -1, 0, 1, 2, \ldots$

WHOLE NUMBERS

W = $0, 1, 2, 3, \ldots$

NATURAL NUMBERS

N = $1, 2, 3, \ldots$

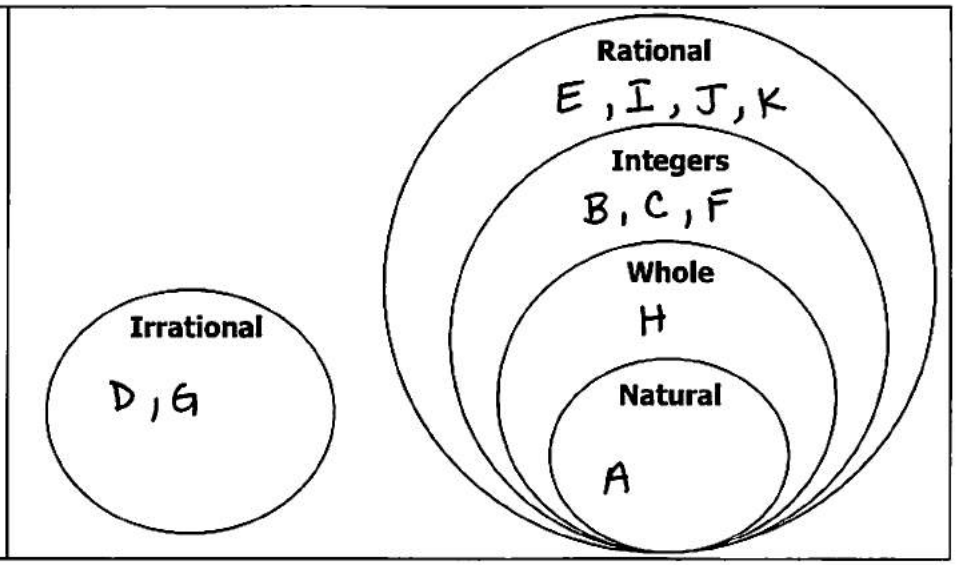
Directions: Name all sets of numbers to which each number belongs.		
1. 30 N, W, Z, Q, R	**2.** -11 Z, Q, R	**3.** $5\frac{4}{7}$ Q, R
4. $\sqrt{21}$ I, R	**5.** 0 W, Z, Q, R	**6.** $-\sqrt{9}$ Z, Q, R
7. $\frac{6}{3}$ N, W, Z, Q, R	**8.** π I, R	**9.** $5.\overline{3}$ Q, R
10. $\frac{\sqrt{144}}{-3}$ Z, Q, R	**11.** -2.875 Q, R	**12.** $\frac{\sqrt{17}}{\sqrt{17}}$ N, W, Z, Q, R

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Organizing the Real Numbers

Directions: Place the LETTER of each value in its location in the real number system below.

A. 7	**B.** $-\frac{36}{12}$
C. $-\sqrt{36}$	**D.** π
E. 6.48	**F.** -2
G. $-\sqrt{10}$	**H.** 0
I. $2\frac{1}{6}$	**J.** $-0.\overline{7}$
K. $\sqrt{\frac{4}{81}}$	



1. Which expression does **not** name an integer?

A. -15 **C.** 0

B. $\frac{6}{14}$ **D.** $\frac{12}{6}$

2. Which expression represents an irrational number?

A. $0.\overline{18}$ **C.** $\frac{2}{3}$

B. $\sqrt{75}$ **D.** $\sqrt{3}-\sqrt{3}$

3. Which number is **not** a whole number?

A. 8 **C.** 0

B. -10 **D.** $\frac{18}{3}$

4. Which of the following is a true statement?

A. -9 is a whole number

B. $\sqrt{25}$ is an irrational number

C. 0 is a natural number

D. $\frac{2}{3}$ is a rational number

5. Which of the following statements is false?

A. All real numbers are rational numbers.

B. Every integer is a rational number.

C. All natural numbers are integers.

D. Every whole number is a real number.

6. Give an example of a number that is a whole number, but not a natural number. 0

7. Give an example of a real number that is not rational. $\sqrt{3}, \pi$ * answers vary

8. Give an example of a rational number that is not an integer. 5.2 *

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Name: _____ **Unit 1:** Algebra Basics

Date: _____ **Bell:** _____ **Homework 1:** The Real Numbers

Directions: Name all sets of numbers to which each real number belongs.

1. 12 N, W, Z, Q, R
2. -15 Z, Q, R
3. $1\frac{1}{2}$ Q, R
4. 3.18 Q, R
5. $\sqrt{48}$ I, R
6. $9.\overline{3}$ Q, R
7. $-2\frac{7}{9}$ Q, R
8. $\sqrt{25}$ N, W, Z, Q, R
9. $\sqrt{3}$ I, R
10. $-\sqrt{64}$ Z, Q, R
11. $-\sqrt{12}$ I, R
12. $\frac{8}{4}$ N, W, Z, Q, R

Directions: Place the LETTER of each value in its location in the real number system below.

A. $2.125\overline{}$ B. 0

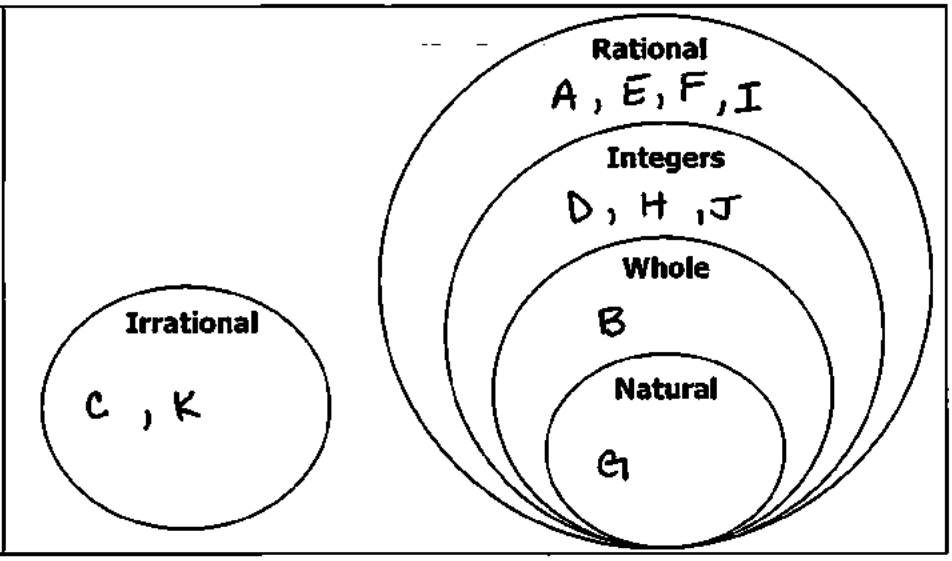
C. π D. $-\frac{15}{5}$

E. $3\frac{1}{4}$ F. $0.\overline{4}$

G. 23 H. -1

I. $\sqrt{\frac{16}{25}}$ J. $-\sqrt{49}$

K. $-\sqrt{35}$



Questions:

1. Give an example of a whole number that is not a natural number. 0
2. Give an example of a rational number that is not a whole number. 5/4 * answers vary
3. Give an example of a real number that is irrational. π *
4. Are integers always/sometimes/never whole numbers? Sometimes
5. Are natural numbers always/sometimes/never rational numbers? always
6. Are irrational numbers always/sometimes/never rational numbers? never

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PROPERTIES OF REAL NUMBERS

In math, properties are statements that are true for any numbers.
They justify steps when simplifying expressions and solving equations.

PROPERTY	MAIN IDEA	EXAMPLES
COMMUTATIVE *Property* (of Addition or Multiplication)	order of values does not matter!	$a+b = b+a$ $2+3 = 3+2$ $x \cdot y = y \cdot x$ $5 \cdot 4 = 4 \cdot 5$
ASSOCIATIVE *Property* (of Addition or Multiplication)	grouping of values does not matter!	$(a+b)+c = a+(b+c)$ $(5+1)+3 = 5+(1+3)$ $(a \cdot b) \cdot c = a \cdot (b \cdot c)$ $(2 \cdot 4) \cdot 7 = 2 \cdot (4 \cdot 7)$
IDENTITY *Property* (of Addition or Multiplication)	Stays the same!	$a+0 = a$ $a \cdot 1 = a$
INVERSE *Property* (of Addition or Multiplication)	Using opposite to "cancel" a value!	$a + (-a) = 0$ $a \cdot \frac{1}{a} = 1$
PROPERTY OF ZERO	Multiplying by 0 always equals 0!	$17 \cdot 0 = 0$ $a \cdot 0 = 0$ $-\frac{1}{a} \cdot 0 = 0$
DISTRUBITVE *Property*	Multiply a value to an expression inside ( ).	$a(b+c) = ab + ac$ $2(x+7) = 2x + 14$ $-5(y-z) = -5y + 5z$

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New PROPERTIES

PROPERTY	MAIN IDEA	EXAMPLES
REFLEXIVE *Property*	A value always equals itself! $a = a$	$16 = 16$ $5x = 5x$ $4y^2 = 4y^2$
SYMMETRIC *Property*	If $a = b$, then $b = a$.	If $x = 3$, then $3 = x$. If $5x = 30$, then $30 = 5x$.
TRANSITIVE *Property*	If $a = b$ and $b = c$, then $a = c$.	If $4 + 3 = 7$ and $7 = \sqrt{49}$, then $4 + 3 = \sqrt{49}$.

CLOSURE *Property*	A set is **closed** (under an operation) if the operation always produces an element of the same set. If an element outside the set is produced, then the operation is **not closed.**
	Directions: Answer True/False. If false, give a counterexample.
	1) Integers are closed under multiplication. True
	2) Natural numbers are closed under subtraction. False $5 - 9 = -4$
	3) Even numbers are closed under addition. True
	4) Irrational numbers are closed under division. False $\frac{\pi}{\pi} = 1$
	5) Perfect squares are closed under subtraction. False $25 - 4 = 21$

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Name: _____ **Unit 1:** Algebra Basics

Date: _____ **Bell:** _____ **Homework 2:** Algebraic Properties

**** This is a 2-page document! ****

1. Commutative Property

Explain: Order doesn't matter with multiplication and addition

Two Examples (NOT from class!)

- $3x+1 = 1+3x$
- $-5y \cdot 4 = 4 \cdot -5y$

2. Associative Property

Explain: Grouping doesn't matter with multiplication and addition

Two Examples (NOT from class!)

- $(8+7)+2 = 8+(7+2)$
- $(5 \cdot 2) \cdot x = 5 \cdot (2 \cdot x)$

3. Identity Property

Explain: Keep the value/ Stays the same

Two Examples (NOT from class!)

- $5x+0 = 5x$
- $6 \cdot 1 = 6$

4. Inverse Property

Explain: Use the opposite to "cancel"

Two Examples (NOT from class!)

- $5 + (-5) = 0$
- $\frac{1}{6} \cdot 6 = 1$

5. Property of Zero

Explain: Anything multiplied by zero equals zero.

Two Examples (NOT from class!)

- $(7x) \cdot 0 = 0$
- $0 \cdot (-3a) = 0$

6. Distributive Property

Explain: Multiplying a value to everything in the parenthesis

Two Examples (NOT from class!)

- $\frac{1}{2}(4x-10) = 2x-5$
- $-6(x+y) = -6x-6y$

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7. Reflexive Property

Explain: A value is equal to itself.

Two Examples (NOT from class!)

- $X = X$
- $5y = 5y$

8. Symmetric Property

Explain: If $a = b$, then $b = a$.

Two Examples (NOT from class!)

- If $2x = 6$, then $6 = 2x$.
- If $5 + 2 = 7$, then $7 = 5 + 2$.

9. Transitive Property

Explain: If $a = b$ and $b = c$, then $a = c$.

One Example (NOT from class!)

- If $5^2 = 25$ and $25 = 24 + 1$, then $5^2 = 24 + 1$.
- If $6x = 12$ and $12 = 3 \cdot 4$, then $6x = 3 \cdot 4$.

10. Use #1-9 to identify each property below.

5 **A.** $ab^2 \cdot 0 = 0$

2 **B.** $(7 + 5) + 1 = 7 + (5 + 1)$

7 **C.** $6 = 6$

6 **D.** $7(x - 3) = 7x - 21$

4 **E.** $x + (-x) = 0$

8 **F.** if $y = 3$, then $3 = y$

1 **G.** $(a + b) + 0 = 0 + (a + b)$

9 **H.** if $x = -1$, and $-1 = z$, then $x = z$

3 **I.** $4x \cdot 1 = 4x$

11. Is the set of integers closed under division? **Explain why or why not.**
If it is not closed, give an example.

No; $\frac{-5}{10} = -\frac{1}{2}$ (not an integer)

12. Is the set of real numbers closed under addition? **Explain why or why not.**
If it is not closed, give an example.

yes!

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Properties WarmUp!!!

G 1. $16 = 16$

B 2. $7 + (a + b) = (7 + a) + b$

F 3. $(4y + 1) \cdot 0 = 0$

I 4. If $5 \cdot 2 = 10$ and $10 = \sqrt{100}$, then $5 \cdot 2 = \sqrt{100}$

C 5. $-2(x - y) = -2x + 2y$

E 6. $3 \cdot \frac{1}{3} = 1$

H 7. If $8x = 24$, then $24 = 8x$

A 8. $(2 + 1) + 9 = (1 + 2) + 9$

D 9. $6m \cdot 1 = 6m$

A. Commutative Property

B. Associative Property

C. Distributive Property

D. Identity Property

E. Inverse Property

F. Property of Zero

G. Reflexive Property

H. Symmetric Property

I. Transitive Property

10. True/False: Negative numbers closed under subtraction. Provide an example if false.

False; $-3 - (-7) = 4$ (not negative)

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Sell THAT PrOPerTY!



Unit 1 Performance Task

Properties of Real Numbers

Currently, math students are struggling to understand the properties of real numbers. This is having an impact on their ability to solve problems. The Math Real Estate Group has hired you to create a poster showing the applications of the real number properties. You will be working in teams and each given one property to display. Your goal is to create a **slogan** for the property, an **explanation** of what the property does, **visuals**, and **at least 4 examples** of the property in action. You want to encourage other students to "buy" your property!

This poster must be creative, unique, and show your understanding of the property.

Task Rubric:	**4** Real Understanding!	**3** Real Close!	**2** "Real" It in!	**1** Not Really!
Mathematical Concepts (Explanation of Property)	Explanation is detailed and clear. Correct terminology and notation are always used.	Explanation is clear. Correct terminology and notation are almost always used.	Explanation is a little difficult to understand. Correct terminology is sometimes used.	Explanation is difficult to understand or is missing. There is little use, or a lot of inappropriate use, of terminology and notation.
Effectiveness	Outstanding use of slogan and visuals to depict importance of property. High degree of effectiveness.	Good use of slogan and visuals to depict importance of property.	Slogans and visuals somewhat depict importance of property.	Slogans and visuals incorrectly depict importance of property OR were not included.
Examples	All example problems are completed and correct.	All but one of the example problems are completed and correct.	All but two of the example problems are completed and correct.	There or more of the example problems are not completed or incorrect.
Neatness and Organization	The work is presented in a neat, clear, organized fashion that is easy to read.	The work is presented in a neat and organized fashion that is usually easy to read.	The work is presented in an organized fashion but may be hard to read at times.	The work appears sloppy and unorganized. It is hard to know what information goes together.

Project Grade: _____

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Name: _____ **Unit 1:** Algebra Basics

Date: _____ **Bell:** _____ **Homework 3:** Quiz 1 Review

Directions: Name all sets of numbers to which each real number belongs.

1. $-\frac{14}{2}$ Z, Q, R	2. $\sqrt{64}$ N, W, Z, Q, R	3. 0 W, Z, Q, R
4. π I, R	5. $0.\overline{45}$ Q, R	6. $\frac{3}{8}$ Q, R

Directions: Place the LETTER of each value in its location in the real number system below.

A. $-0.\bar{2}$ B. 18

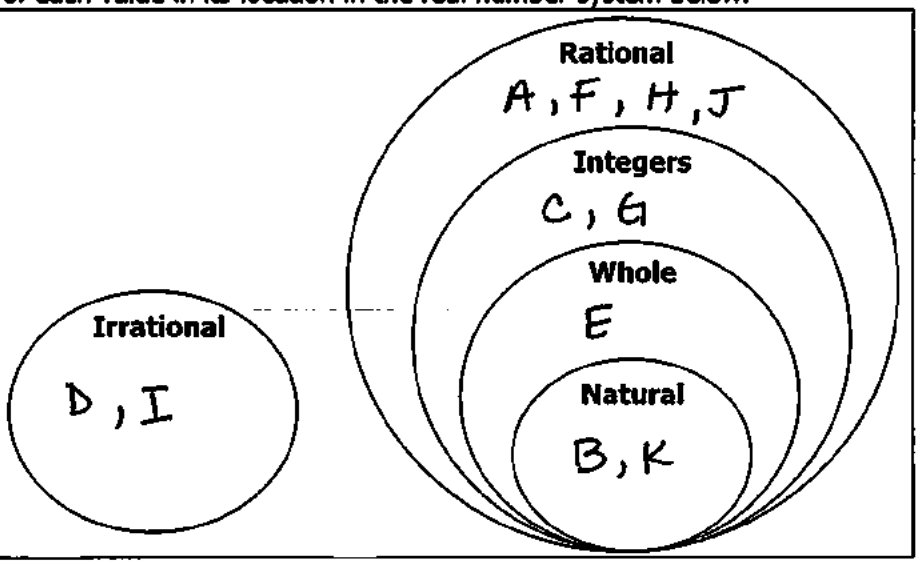
C. $-\sqrt{100}$ D. π

E. 0 F. $2\frac{1}{6}$

G. -5 H. 4.03

I. $-\sqrt{50}$ J. $\sqrt{\frac{4}{9}}$

K. $\frac{36}{9}$



Directions: Name the property shown.

1. $4 + (x + y) = (4 + x) + y$ Assoc. Property	6. $2\left(\frac{1}{2}\right) = 1$ Inverse Property
2. if $\sqrt{49} = 7$ and $7 = 3 + 4$, then $\sqrt{49} = 3 + 4$ Transitive Property	7. $3 = 3$ Reflexive Property
3. $(mn) \cdot 1 = mn$ Identity Property	8. $10k^2 + (-10k^2) = 0$ Inverse Property
4. $(x+3) \cdot 0 = 0$ Zero Property	9. $5(x + 7) = 5x + 35$ Distributive Property
5. $(x + 2) + y = (2 + x) + y$ Commutative Property	10. if $-7v = 28$, then $28 = -7v$ Symmetric Property

***Also, make sure you know the closure property!**

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Name: _____ **Algebra I**

Date: _____ **Bell:** _____ **Unit 1:** Algebra Basics

Quiz 1-1: The Real Numbers & Properties

For Questions 1 – 5: Put an X in each set the number belongs.

	R (Real)	**I** (Irrational)	**Q** (Rational)	**Z** (Integers)	**W** (Whole)	**N** (Natural)
1. 10	X		X	X	X	X
2. $\frac{5}{6}$	X		X			
3. $\sqrt{24}$	X	X				
4. 0	X		X	X	X	
5. $-\sqrt{81}$	X		X	X		

6. Place the LETTER of each value in its location in the real number system below.

A. $\sqrt{0}$ **B.** $\sqrt{48}$

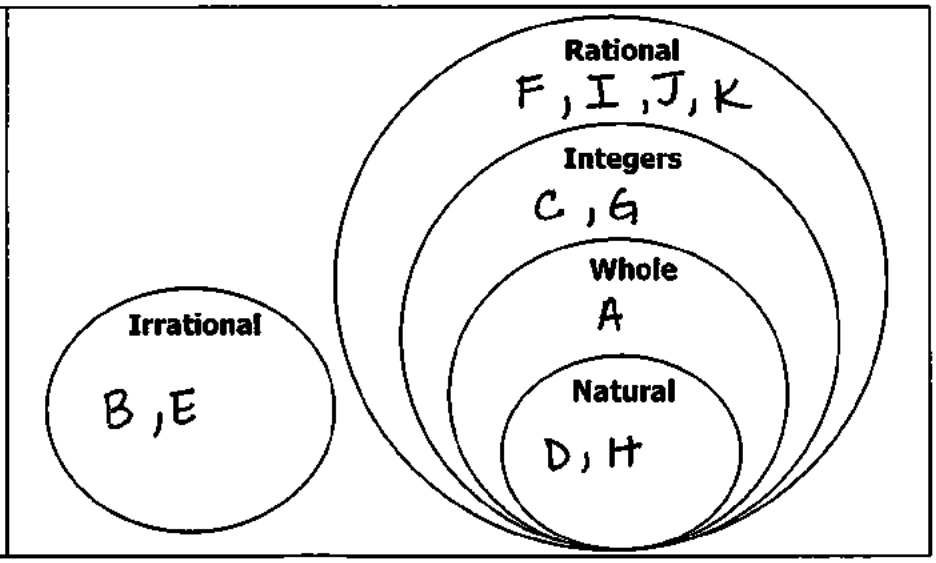
C. $-\frac{24}{8}$ **D.** 17

E. π **F.** -3.75

G. -1 **H.** $\sqrt{36}$

I. $\sqrt{\frac{16}{49}}$ **J.** $0.\overline{9}$

K. $2\frac{1}{5}$



Directions: Circle the best answer.

7. Which does **not** represent an integer? **A.** 0 **B.** $-\frac{18}{2}$ **C.** $\sqrt{9}$ **D.** $\frac{4}{5}$

8. Which is an irrational number? **A.** $\sqrt{21}$ **B.** $3\frac{5}{8}$ **C.** $2.\overline{41}$ **D.** -13

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9. Which statement is true?

A. $\sqrt{81}$ is an irrational number

B. 0 is a natural number

C. $4.\bar{6}$ is a rational number

D. -9 is a whole number

10. Which statement is false?

A. All whole numbers are integers.

B. All real numbers are irrational numbers.

C. All natural numbers are whole numbers.

D. All whole numbers are real numbers.

For Questions 11 – 18: Identify the property that justifies the statement.
Use the property bank below for spelling!

11. If $y = -5$, then $-5 = y$ Symmetric	**12.** $8a + 4 = 4 + 8a$ Commutative
13. $12(x + 4) = 12x + 48$ Distributive	**14.** $n^2 \cdot 0 = 0$ Zero
15. $8 + (2 + 6) = (8 + 2) + 6$ Associative	**16.** If $3^2 = 9$ and $9 = \sqrt{81}$, then $3^2 = \sqrt{81}$ Transitive
17. $-7 = -7$ Reflexive	**18.** $6s \cdot \frac{1}{6s} = 1$ Inverse

Property Bank:

Commutative
Associative
Distributive
Identity
Inverse
Property of Zero
Reflexive
Symmetric
Transitive

19. Which set of numbers is closed under subtraction?

A. Whole Numbers $5-6=-1$

B. Natural Numbers $2-9=-7$

C. Integers

D. Irrational Numbers $\pi-\pi=0$

20. Which of the following sets is NOT closed under addition?

A. Natural Numbers

B. Whole Numbers

C. Integers

D. Odd Numbers $3+7=10$

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Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples
Absolute Value	The distance from zero on a number line

Examples

Directions: Evaluate each expression.

1. $\lvert -13 \rvert =$ 13	**2.** $\lvert 21 \rvert =$ 21	**3.** $\lvert -3 \rvert + \lvert -5 \rvert =$ $3+5=8$
4. $\lvert 9 \rvert + \lvert -8 \rvert =$ $9+8=17$	**5.** $\lvert -12 \rvert + \lvert 15 \rvert =$ $12+15=27$	**6.** $\lvert 21-17 \rvert =$ $\lvert 4 \rvert = 4$
7. $\lvert -11 \rvert - \lvert -5 \rvert =$ $11-5=6$	**8.** $\lvert 4 \rvert - \lvert -4 \rvert =$ $4-4=0$	**9.** $\lvert 15-23 \rvert =$ $\lvert -8 \rvert = 8$
10. $\lvert 24 \rvert - \lvert -15 \rvert =$ $24-15=9$	**11.** $\lvert 28 \rvert - \lvert -26 \rvert =$ $28-26=2$	**12.** $\lvert -3 \rvert - \lvert 19 \rvert =$ $3-19=-16$

Order of Operations

Simplifying an expression using a specific order

P	Parenthesis + Grouping Symbols
E	Exponents
M/D	Multiplication / Division from left → right
A/S	Addition / Subtraction from left → right

Examples

Directions: Evaluate each expression.

13. $7+54\div 3(2)$

$= 7+18(2)$

$= 7+36$

$= 43$

14. $17-2^5\div 4+6$

$= 17-32\div 4+6$

$= 17-8+6$

$= 9+6$

$= 15$

15. $24-4^2\cdot 3+15$

$= 24-16\cdot 3+15$

$= 24-48+15$

$= -24+15$

$= -9$

16. $\dfrac{(7-2^2)+17}{-14+2\cdot 5}$

$= \dfrac{(7-4)+17}{-14+10}$

$= \dfrac{20}{-4} = -5$

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17. $(8-5)^3-|5^2-4^3|\div 3$

$= (3)^3 - |25-64| \div 3$

$= 27 - |-39| \div 3$

$= 27 - 13$

$= \boxed{14}$

18. $|-7^2+3^2\cdot 4|+18\div 2\cdot 5$

$= |-49+9\cdot 4| + 9\cdot 5$

$= |-49+36| + 45$

$= |-13| + 45$

$= 13 + 45 = \boxed{58}$

19. $\dfrac{5^2-3^4}{(2^3-5)\cdot(6^2\div 9)}$

$= \dfrac{25-81}{(8-5)\cdot(36\div 9)}$

$= \dfrac{-56}{3\cdot 4} = \dfrac{-56}{12} = \boxed{\dfrac{-14}{3}}$

20. $-|-9-10|+12^2\div 6\cdot 3$

$= -|-19| + 144 \div 6\cdot 3$

$= -19 + 24\cdot 3$

$= -19 + 72$

$= \boxed{53}$

21. $20^2\div\{3(5-9)^2+2\}$

$= 400 \div \{3(-4)^2+2\}$

$= 400 \div \{3(16)+2\}$

$= 400 \div \{48+2\}$

$= 400 \div 50$

$= \boxed{8}$

22. $\dfrac{12\left[30-(9+4^2)\right]}{|10|-|-6|}$

$= \dfrac{12[30-(9+16)]}{10-6}$

$= \dfrac{12[30-25]}{4}$

$= \dfrac{12(5)}{4} = \dfrac{60}{4} = \boxed{15}$

23. $\dfrac{84}{12}+(5-7^2)-72\div 6\cdot 2$

$= 7+(5-49)-12\cdot 2$

$= 7+(-44)-24$

$= -37-24$

$= \boxed{-61}$

24. $\dfrac{-(8-11)^3-3|-14|}{2^4-7}$

$= \dfrac{-(-3)^3-3(14)}{16-7}$

$= \dfrac{-(-27)-42}{9}$

$= \dfrac{27-42}{9} = \dfrac{-15}{9} = \boxed{\dfrac{-5}{3}}$

Summary: _____

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Name: _____ **Unit 1:** Algebra Basics

Date: _____ **Bell:** _____ **Homework 4:** Order of Operations

Directions: Evaluate each expression using the order of operations.

1. $17-5\cdot4\div2$

$17-20\div2$

$17-10 = \boxed{7}$

2. $40-32\div8+5\cdot-2$

$40-4+(-10)$

$36-10 = \boxed{26}$

3. $35-14\div2+8^2$

$35-7+64$

$28+64 = \boxed{92}$

4. $(4-7)^2-6\cdot7+20$

$(-3)^2-42+20$

$9-42+20$

$-33+20 = \boxed{-13}$

5. $2[18-(5+3^2)\div7]$

$2[18-(5+9)\div7]$

$2[18-14\div7]$

$2[18-2] = 2(16) = \boxed{32}$

6. $[(-5+1)\div2]^3-|-7|$

$[(-4)\div2]^3-7$

$[-2]^3-7$

$-8-7 = \boxed{-15}$

7. $1+(-2-5)^2+(14-17)\cdot4$

$1+(-7)^2+(-3)\cdot4$

$1+49+(-12)$

$50-12 = \boxed{38}$

8. $\frac{6(2+4)-1}{2\cdot3+1}$

$\frac{6(6)-1}{6+1} = \frac{36-1}{7} = \frac{35}{7}$

$= \boxed{5}$

9. $\frac{2^5\div8+3}{4+3}$

$\frac{32\div8+3}{7} = \frac{4+3}{7} = \frac{7}{7}$

$= \boxed{1}$

10. $\frac{17\cdot5-3\cdot5}{3^2+1}$

$\frac{85-15}{9+1} = \frac{70}{10} = \boxed{7}$

11. $\frac{13+7^2\div7}{9-20\div4-16}$

$\frac{13+49\div7}{9-5-16} = \frac{13+7}{4-16} = \frac{20}{-12}$

$= \boxed{-\frac{5}{3}}$

12. $-|-16|+\frac{(3^2-4)}{5-10}$

$-16\div\frac{9-4}{-5} = -16\div\frac{5}{-5}$

$= -16\div-1$

$= \boxed{16}$

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Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples
ALGEBRAIC EXPRESSION	A mathematical phrase made up of numbers and variables.
SUBSTITUTION *Property*	If $a = b$, then a may be replaced for b in any expression
EVALUATING *Expressions*	**To evaluate an expression variable replacements:** • Substitute the variables with their given values. • Each time you substitute a variable with a number, put parenthesis around the number! • Follow the order of operations to evaluate!
EXAMPLES	**Directions:** Evaluate each expression using the variable replacements.

1. $ab^2 + c$ if $a = 2$, $b = 4$, $c = 7$

$(2)(4)^2 + 7$

$2(16) + 7$

$32 + 7 = \boxed{39}$

2. $3x^2 - 4x$ if $x = -2$

$3(-2)^2 - 4(-2)$

$3(4) + 8$

$12 + 8 = \boxed{20}$

YOU TRY!	**Directions:** Evaluate each expression using the variable replacements.

3. $a^2b - b^2$ if $a = 3$ and $b = -4$

$(3)^2(-4) - (-4)^2$

$9(-4) - (16)$

$-36 - 16 = \boxed{-52}$

4. $a^2b - b^2$ if $a = 4$ and $b = -7$

$(4)^2(-7) - (-7)^2$

$16(-7) - 49$

$-112 - 49 = \boxed{-161}$

5. $-y^2 - 3xy$ if $x = -4$ and $y = 2$

$-(2)^2 - 3(-4)(2)$

$-4 + 12(2)$

$-4 + 24 = \boxed{20}$

6. $-y^2 - 3xy$ if $x = -\frac{5}{6}$ and $y = -12$

$-(-12)^2 - 3(-5/6)(-12)$

$-144 + (5/2)(-12)$

$-144 - 30 = \boxed{-174}$

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7. $\frac{a^2+b^2}{a-b}$ if $a = 3$ and $b = 4$

$\frac{(3)^2+(4)^2}{3-4}$

$\frac{9+16}{-1} = \frac{25}{-1} = \boxed{-25}$

8. $\frac{a^2+b^2}{a-b}$ if $a = -3$ and $b = -5$

$\frac{(-3)^2+(-5)^2}{(-3)-(-5)}$

$\frac{9+25}{-3+5} = \frac{34}{2} = \boxed{17}$

9. $x^2 - 2(x-y) - z^3$ if $x = 2$, $y = -2$, and $z = -3$

$(2)^2-2(2-(-2)) - (-3)^3$

$4-2(4) - (-27)$

$4-8+27$

$-4+27 = \boxed{23}$

10. $x^2 - 2(x-y) - z^3$ if $x = -3$, $y = 4$ and $z = -1$

$(-3)^2-2(-3-4) - (-1)^3$

$9-2(-7) - (-1)$

$9+14+1$

$23+1 = \boxed{24}$

11. $-a^2 - 2bc - |c|$ if $a = -2$, $b = 3$, and $c = -3$

$-(-2)^2-2(3)(-3) - |-3|$

$-4-6(-3)-3$

$-4+18-3$

$14-3 = \boxed{11}$

12. $-a^2 - 2bc - |c|$ if $a = -3$, $b = -5$, and $c = 2$

$-(-3)^2-2(-5)(2) - |2|$

$-9+10(2)-2$

$-9+20-2$

$11-2 = \boxed{9}$

Summary: _____

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Name: _____ **Unit 1:** Algebra Basics

Date: _____ **Bell:** _____ **Homework 5:** Evaluating Expressions

Directions: Evaluate each expression for the given variable values.

1. $n^2 - 3n + 8$ if $n = 4$

$(4)^2 - 3(4) + 8$

$16 - 12 + 8$

$4 + 8 = \boxed{12}$

2. $\frac{x^2 - 4y}{2}$ if $x = 4$ and $y = -3$

$\frac{(4)^2 - 4(-3)}{2} = \frac{16 + 12}{2} = \frac{28}{2}$

$= \boxed{14}$

3. $4|m - n|$ if $m = -7$ and $n = 2$

$4|(-7) - 2|$

$4|-9|$

$4(9) = \boxed{36}$

4. $19 - x^2$ if $x = -5$

$19 - (-5)^2$

$19 - 25 = \boxed{-6}$

5. $s^2t - 10$ if $s = -8$ and $t = \frac{3}{4}$

$(-8)^2(3/4) - 10$

$64(3/4) - 10$

$48 - 10 = \boxed{38}$

6. $4p^2 + 7q^3$ if $p = -3$ and $q = -2$

$4(-3)^2 + 7(-2)^3$

$4(9) + 7(-8)$

$36 - 56 = \boxed{-20}$

7. $\frac{(a+b)^5}{a - 2b^2}$ if $a = 6$ and $b = -8$

$\frac{(6 + (-8))^5}{6 - 2(-8)^2} = \frac{(-2)^5}{6 - 2(64)} = \frac{-32}{6 - 128}$

$= \frac{-32}{-122} = \boxed{\frac{16}{61}}$

8. $\frac{5c^2 - d^2 + 3}{2c - 4d}$ if $c = -1$ and $d = -4$

$\frac{5(-1)^2 - (-4)^2 + 3}{2(-1) - 4(-4)} = \frac{5 - 16 + 3}{-2 + 16}$

$= \frac{-8}{14} = \boxed{-\frac{4}{7}}$

9. $-q^2 - r^2 + 3s$ if $q = 9$, $r = -6$, $s = -20$

$-(9)^2 - (-6)^2 + 3(-20)$

$-81 - 36 - 60$

$-117 - 60 = \boxed{-177}$

10. $3|x + y|^2 - (xy)^2$ if $x = 3$ and $y = -5$

$3|3 + (-5)|^2 - (3 \cdot -5)^2$

$3|-2|^2 - (-15)^2$

$3(4) - 225$

$12 - 225 = \boxed{-213}$

11. $2x^2 - 5xy - y^3$ if $x = -3$ and $y = -2$

$2(-3)^2 - 5(-3)(-2) - (-2)^3$

$2(9) + 15(-2) - (-8)$

$18 - 30 + 8$

$-12 + 8 = \boxed{-4}$

12. $-a^2 + 7b^4 - 2c^3$ if $a = -4$, $b = -2$, $c = -3$

$-(-4)^2 + 7(-2)^4 - 2(-3)^3$

$-16 + 7(16) - 2(-27)$

$-16 + 112 + 54$

$96 + 54 = \boxed{150}$

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Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples
MATRICES	A rectangular array of data in rows + columns.
ELEMENTS	The individual numbers in the matrix.
ROWS	The horizontal elements (left → right)
COLUMNS	The vertical elements (top → bottom)
DIMENSIONS	Identifies the matrix by the # of rows and columns, with rows stated first
EXAMPLE	**Sue and Eva made a table of their science quiz grades:**

Sue	B	C	A	B	A	B	A	C	B	A	B	C
Eva	A	B	B	A	A	C	B	B	A	A	B	B

1. Create a matrix to organize the data:

$$\begin{matrix} & A & B & C \\ \text{Sue} & 4 & 5 & 3 \\ \text{Eva} & 5 & 6 & 1 \end{matrix}$$

2. Give the dimensions: 2 x 3

IDENTIFYING ELEMENTS	Element $X_{(a,b)}$ is used to identify a specific element was a is the row number and b is the column number.
EXAMPLES	**Directions:** Given matrix M to the left, identify each element.

$$M = \begin{bmatrix} 5 & 3 & 1 & 0 \\ -2 & -8 & 2 & 11 \\ 7 & 9 & 1 & -5 \end{bmatrix}$$

3. Element $X_{(1,2)}$ = 3
4. Element $X_{(3,2)}$ = 9
5. Element $X_{(2,4)}$ = 11
6. Element $X_{(1,3)}$ = 1
7. Element $X_{(3,4)}$ = -5
8. Element $X_{(2,1)}$ = -2
9. Element $X_{(3,1)}$ = 7
10. Element $X_{(2,2)}$ = -8
11. What are the dimensions of matrix M? 3 x 4

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ADDITION & SUBTRACTION

- Matrices must have the same dimensions in order to add or subtract them.
- Simply add or subtract corresponding elements to create the new matrix.

EXAMPLES

Using the matrices below, perform each operation, if possible.

$A = \begin{bmatrix} 7 & 19 \\ 4 & -6 \\ -1 & -2 \end{bmatrix}$ $B = \begin{bmatrix} -12 & 9 \\ -5 & -1 \\ 16 & 2 \end{bmatrix}$ $C = \begin{bmatrix} 1 & 13 & -4 \\ 2 & -8 & 7 \end{bmatrix}$ $D = \begin{bmatrix} -14 & 0 \\ 1 & -5 \\ 10 & 9 \end{bmatrix}$

12. $B + D$

$\begin{bmatrix} -26 & 9 \\ -4 & -6 \\ 26 & 11 \end{bmatrix}$

13. $B - A$

$\begin{bmatrix} -19 & -10 \\ -9 & 5 \\ 17 & 4 \end{bmatrix}$

14. $C + B$

No Solution!

15. $A - D$

$\begin{bmatrix} 21 & 19 \\ 3 & -1 \\ -11 & -11 \end{bmatrix}$

SCALAR MULTIPLICATION

A matrix can be multiplied by a scalar (any real number) by multiplying the scalar to each separate element.

EXAMPLES

Using the matrices given above, perform each operation.

16. $4C$

$\begin{bmatrix} 4 & 52 & -16 \\ 8 & -32 & 28 \end{bmatrix}$

17. $-2A$

$\begin{bmatrix} -14 & -38 \\ -8 & 12 \\ 2 & 4 \end{bmatrix}$

18. $3D + B$

$\begin{bmatrix} -42 & 0 \\ 3 & -15 \\ 30 & 21 \end{bmatrix} + \begin{bmatrix} -12 & 9 \\ -5 & -1 \\ 16 & 2 \end{bmatrix}$

$= \begin{bmatrix} -54 & 9 \\ -2 & -16 \\ 46 & 29 \end{bmatrix}$

19. $5D - 2B$

$\begin{bmatrix} -70 & 0 \\ 5 & -25 \\ 50 & 45 \end{bmatrix} - \begin{bmatrix} -24 & 18 \\ -10 & -2 \\ 32 & 4 \end{bmatrix}$

$= \begin{bmatrix} -46 & -18 \\ 15 & -23 \\ 18 & 41 \end{bmatrix}$

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Name: _____ **Unit 1:** Algebra Basics

Date: _____ **Bell:** _____ **Homework 6:** Matrices

Directions: State the dimensions of each matrix.

1. $\begin{bmatrix} 3 & -4 & -9 \\ 2 & -7 & 0 \end{bmatrix}$ 2x3

2. $\begin{bmatrix} 20 \\ -6 \end{bmatrix}$ 2x1

3. $\begin{bmatrix} 10 & 7 \\ 0 & 16 \\ -3 & -8 \\ -5 & 1 \end{bmatrix}$ 4x2

4. $\begin{bmatrix} 10 & -4 & -20 & 15 \end{bmatrix}$ 1x4

Directions: Add/Subtract the following matrices (if possible)

5. $\begin{bmatrix} -2 & 6 \\ 1 & 4 \end{bmatrix} + \begin{bmatrix} -3 & 11 \\ 2 & 7 \end{bmatrix}$

$\begin{bmatrix} -5 & 17 \\ 3 & 11 \end{bmatrix}$

6. $\begin{bmatrix} 14 & -3 & 1 \end{bmatrix} - \begin{bmatrix} 2 & -4 & -7 \end{bmatrix}$

$\begin{bmatrix} 12 & 1 & 8 \end{bmatrix}$

7. $\begin{bmatrix} 2 & -5 \\ 6 & 8 \end{bmatrix} + \begin{bmatrix} 10 \\ 15 \end{bmatrix}$

No Solution!

8. $\begin{bmatrix} 13 & -10 & 12 \\ 6 & 3 & 30 \\ 0 & -5 & 5 \end{bmatrix} - \begin{bmatrix} 1 & 6 & -7 \\ 2 & -5 & 8 \\ -3 & 4 & 9 \end{bmatrix}$

$\begin{bmatrix} 12 & -16 & 19 \\ 4 & 8 & 22 \\ 3 & -9 & -4 \end{bmatrix}$

Directions: Use scalar multiplication to evaluate the following matrices.

9. $-3\begin{bmatrix} 12 & -5 \end{bmatrix}$

$\begin{bmatrix} -36 & 15 \end{bmatrix}$

10. $8\begin{bmatrix} -1 & 2 \\ 0 & 4 \end{bmatrix}$

$\begin{bmatrix} -8 & 16 \\ 0 & 32 \end{bmatrix}$

Directions: If $A = \begin{bmatrix} 7 & 3 \\ -4 & -1 \end{bmatrix}$, $B = \begin{bmatrix} 9 & 6 \\ 3 & 10 \end{bmatrix}$, and $C = \begin{bmatrix} 0 & -11 \\ 2 & 8 \end{bmatrix}$, find the following:

11. 4A – 2B

$\begin{bmatrix} 28 & 12 \\ -16 & -4 \end{bmatrix} - \begin{bmatrix} 18 & 12 \\ 6 & 20 \end{bmatrix}$

$= \begin{bmatrix} 10 & 0 \\ -22 & -24 \end{bmatrix}$

12. 2C – A

$\begin{bmatrix} 0 & -22 \\ 4 & 16 \end{bmatrix} - \begin{bmatrix} 7 & 3 \\ -4 & -1 \end{bmatrix}$

$= \begin{bmatrix} -7 & -25 \\ 8 & 17 \end{bmatrix}$

13. 3A + B – C

$\begin{bmatrix} 21 & 9 \\ -12 & -3 \end{bmatrix} + \begin{bmatrix} 9 & 6 \\ 3 & 10 \end{bmatrix} - \begin{bmatrix} 0 & -11 \\ 2 & 8 \end{bmatrix}$

$= \begin{bmatrix} 30 & 26 \\ -11 & -1 \end{bmatrix}$

14. B – A

$\begin{bmatrix} 2 & 3 \\ 7 & 11 \end{bmatrix}$

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Name: _____ **Algebra I**

Date: _____ **Bell:** _____ **Unit 1:** Algebra Basics

Quiz 1-2: Order of Operations, Matrices, & Evaluating Expressions

Part I: Absolute Value & Order of Operations Show all work for those with several steps!

1. $|6| + |-2|$

$6 + 2$

2. $2^3 - 5 \cdot (6 + 4) \div 2$

$8 - 5(10) \div 2$

$8 - 50 \div 2$

$8 - 25$

3. $86 - \left[(4-9)^2 \cdot 3\right]$

$86 - \left[(-5)^2 \cdot 3\right]$

$86 - [25 \cdot 3]$

$86 - 75$

Answers

1. 8
2. -17
3. 11
4. 10
5. 2
6. -3

4. $|2 + 3(-4)|$

$|2 - 12|$

$|-10|$

5. $\frac{36 - 3 \cdot 4}{15 - 9 \div 3}$

$\frac{36 - 12}{15 - 3}$

$\frac{24}{12}$

6. $\frac{1 + 3 \cdot 5}{18 - 4^2} - |-11|$

$\frac{1 + 15}{18 - 16} - 11$

$\frac{16}{2} - 11$

$8 - 11$

Part II: Matrices If $A = \begin{bmatrix} 7 & -2 & 3 \\ -4 & 11 & 0 \end{bmatrix}$, $B = \begin{bmatrix} 15 \\ -6 \end{bmatrix}$, $C = \begin{bmatrix} -9 \\ 24 \end{bmatrix}$ and $D = \begin{bmatrix} -6 & 12 & 8 \\ 0 & -1 & 17 \end{bmatrix}$ find:

7. $A + D$

$\begin{bmatrix} 1 & 10 & 11 \\ -4 & 10 & 17 \end{bmatrix}$

8. $C - B$

$\begin{bmatrix} -24 \\ 30 \end{bmatrix}$

9. $B + D$

No solution!

10. $-3B$

$\begin{bmatrix} -45 \\ 18 \end{bmatrix}$

11. $2A - D$

$\begin{bmatrix} 14 & -4 & 6 \\ -8 & 22 & 0 \end{bmatrix} - \begin{bmatrix} -6 & 12 & 8 \\ 0 & -1 & 17 \end{bmatrix}$

$= \begin{bmatrix} 20 & -16 & -2 \\ -8 & 23 & -17 \end{bmatrix}$

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Part III: Evaluating Expressions If $w = -1$, $x = 6$, and $y = -3$, find the following:

12. $2y - 5w$

$2(-3) - 5(-1)$

$-6 + 5$

13. $-\frac{7}{3}x + y^2$

$-\frac{7}{3}(6) + (-3)^2$

$-14 + 9$

14. $3w^2 - 5w + 1$

$3(-1)^2 - 5(-1) + 1$

$3(1) + 5 + 1$

$3 + 5 + 1$

15. $2|y| - x^2$

$2|-3| - (6)^2$

$2(3) - 36$

$6 - 36$

16. $-x^2 - 7xy + w^3$

$-(6)^2 - 7(6)(-3) + (-1)^3$

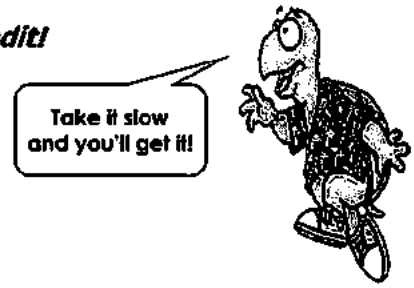
$-36 - 42(-3) - 1$

$-36 + 126 - 1$

$90 - 1$

Answers	
12.	-1
13.	-5
14.	9
15.	-30
16.	89
B.	-14

B. $7\left[4^2 - 6|5 + (2^3 - 18) + 16 \div 8|\right]$ ***Show all steps to receive credit!***



$7\left[16 - 6|5 + (8 - 18) + 16 \div 8|\right]$

$7\left[16 - 6|5 - 10 + 16 \div 8|\right]$

$7\left[16 - 6|-5 + 2|\right]$

$7\left[16 - 6|-3|\right]$

$7[16 - 6(3)]$

$7[16 - 18] \quad = 7(-2) \quad = -14$

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Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples

Parts of an Expression

$$7x + 3 - 5x + 8 + 4x - 1$$

Variable Terms
(Terms WITH a variable)
$7x, -5x, 4x$

Constant Terms
(Terms WITHOUT a variable)
$3, 8, -1$

Coefficients
(Number NEXT TO a variable)
$7, -5, 4$

Combining Like Terms

You can simplify an algebraic expression by **combining like terms.** This means to combine common variable terms and contant terms.

Example: Simplify the expression below:

$7x + 3 - 5x + 8 + 4x - 1 =$ $6x + 10$

Examples

Directions: Simplify each expression.

1. $3x + 6x$ $9x$	**2.** $9a - 10a$ $-a$
3. $7m - 5m - 6$ $2m - 6$	**4.** $9 - 6w + 5$ $-6w + 14$
5. $7p - 1 - 9p + 5$ $-2p + 4$	**6.** $5h - 6 - 8 + 7h$ $12h - 14$
7. $5x + 7y - x + 2y + 3x$ $7x + 9y$	**8.** $26a + 2b - 4a + 10b$ $22a + 12b$

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with Exponents

Terms with the same exponents or combination of variables and exponents are considered like terms. When combined, be sure to NOT change their exponents!

Examples

9. $5x^2 + 7x^2$

$12x^2$

10. $-15n^5 + 4n^5$

$-11n^5$

11. $8c^3 - 7c^3 + 16c^3$

$17c^3$

12. $8k^2 - k - 5k + 7 - 2k^2$

$6k^2 - 6k + 7$

13. $-4a^2b + 7 + 10a^2b$

$6a^2b + 7$

14. $x^2 + 2xy - 7xy + 4y^2$

$x^2 - 5xy + 4y^2$

15. $a^2 + 7ab - 10ab + b^2$

$a^2 - 3ab + b^2$

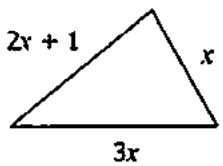
16. $4x^2 - 7 + 2x - 3x^2 + 8x - 6$

$x^2 + 10x - 13$

Geometric Applications

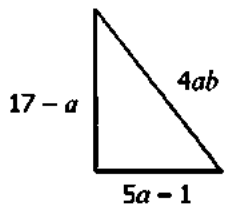
Directions: Give the perimeter of each figure as a simplified expression.

17.



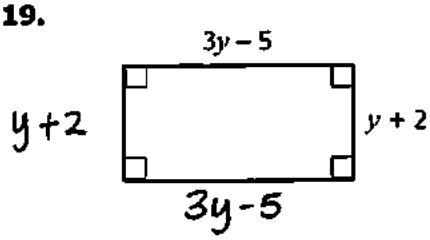
$6x + 1$

18.



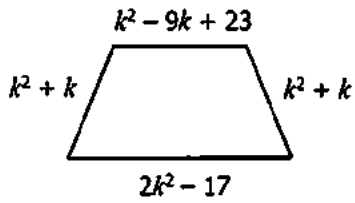
$4a + 4ab + 16$

19.



$8y - 6$

20.



$5k^2 - 7k + 6$

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Name: _____ **Unit 1:** Algebra Basics

Date: _____ **Bell:** _____ **Homework 7:** Combining Like Terms

Directions: Simplify each expression by combining like terms.

1. $8y - 9y$

$-y$

2. $3x - 7x + 1$

$-4x+1$

3. $8a - 6 + a - 1$

$9a-7$

4. $-2y + 14 - 9y + 5$

$-11y+19$

5. $-x - 2 + 15x$

$14x-2$

6. $-7w - 5 - 7w + 9$

$-14w+4$

7. $8d - 4 - d - 2$

$7d-6$

8. $40 - 9x + 4x + 5$

$-5x+45$

9. $9a + 8 - 2a - 3 - 5a$

$2a+5$

10. $4xy - 6xy + 5y - 9x$

$-2xy+5y-9x$

11. $7x + 2y + 5 - 5x + 9y - 1$

$2x+11y+4$

12. $-5f - 7g + f - 5g - 4 + f$

$-3f-12g-4$

13. $-6 + 3a - b - 4a + 4b + 5$

$-a+3b-1$

14. $6n^2 - 2n - 3 + 5n^2 - 9n - 6$

$11n^2-11n-9$

15. $8c^2 - c + 3 + 2c^2 - c + 2$

$10c^2-2c+5$

16. $-12x^2y - 3x^2y$

$-15x^2y$

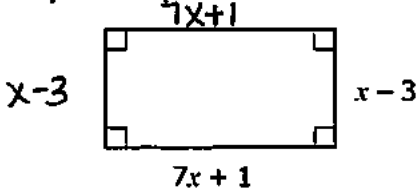
17. $6a^2b^2 + 3ab^2 - a^2b^2 - 4ab^2$

$5a^2b^2-ab^2$

18. $19x^2y - 4xy^2 + 5xy^2 - 16x^2y$

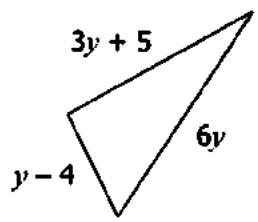
$3x^2y+xy^2$

19. Write the perimeter of the rectangle as a simplified expression.



$16x-4$

20. Write the perimeter of the triangle as a simplified expression.



$10y+1$

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Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples		
Distributive Property Recall: $a(b+c)=$ $ab+ac$ or $a(b-c)=$ $ab-ac$	**1.** $7(x+4)$ $7x+28$	**2.** $2(b-3)$ $2b-6$	**3.** $-4(y+3)$ $-4y-12$
	4. $-5(m-2)$ $-5m+10$	**5.** $-(y-9)$ $-y+9$	**6.** $8(4-b)$ $32-8b$
	7. $-4(-w-10)$ $4w+40$	**8.** $5(2m-3)$ $10m-15$	**9.** $-3(2x-4)$ $-6x+12$
	10. $3(m+n)$ $3m+3n$	**11.** $x(y+4)$ $xy+4x$	**12.** $d(c-4)$ $dc-4d$
	13. $6a(b+2c)$ $6ab+12ac$	**14.** $2(x-y+8)$ $2x-2y+16$	**15.** $x(2y-5z+1)$ $2xy-5xz+x$
	16. $\frac{1}{4}(8x-4)$ $2x-1$	**17.** $\frac{1}{6}\left(\frac{3}{5}x+18\right)$ $\frac{1}{10}x+3$	**18.** $-\frac{1}{8}\left(\frac{4}{5}x-24\right)$ $-\frac{1}{10}x+3$
Distribute AND Combine!	To **simplify an expression** means to ensure there are **no parentheses** and **no like terms**. In order to do this, distribute first (if needed), then combine like terms.		
Examples	**19.** $8(2x-3)-6x$ $16x-24-6x$ $=\boxed{10x-24}$	**20.** $9(2k-4)-2(7k-12)$ $18k-36-14k+24$ $=\boxed{4k-12}$	
	21. $10-(y-6)-y$ $10-y+6-y$ $=\boxed{-2y+16}$	**22.** $6+8(4w-7)-(2w+1)$ $6+32w-56-2w-1$ $=\boxed{30w-51}$	
	23. $\frac{9}{4}\left(\frac{10}{3}x-4\right)-\frac{1}{4}(2x-8)$ $\frac{15}{2}x-9-\frac{1}{2}x+2$ $=\boxed{7x-7}$	**24.** $\frac{7}{8}\left(\frac{4}{5}x+16\right)+\frac{3}{5}\left(\frac{1}{2}x-10\right)$ $\frac{7}{10}x+14+\frac{3}{10}x-6$ $=\boxed{x+8}$	

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Name: _____ **Unit 1:** Algebra Basics

Date: _____ **Bell:** _____ **Homework 8:** Simplifying Expressions

Directions: Simplify each expression by distributing.	
1. $8(x + 5)$ = $8x + 40$	**2.** $3(2y - 7)$ = $6y - 21$
3. $-2(3m + 9)$ = $-6m - 18$	**4.** $-6(z - 1)$ = $-6z + 6$
5. $-(5x + 1)$ = $-5x - 1$	**6.** $-(x + 4)$ = $-x - 4$
7. $b(a - 6)$ = $ab - 6b$	**8.** $7a(3b - 2c + 4)$ = $21ab - 14ac + 28a$
9. $\frac{1}{2}(8x + 2)$ = $4x + 1$	**10.** $\frac{2}{3}\left(\frac{1}{4}x - 6\right)$ = $\frac{1}{6}x - 4$

Directions: Simplify each expression by distributing and combining like terms	
11. $-7(w - 4) + 3w - 27$ $-7w + 28 + 3w - 27$ = $\boxed{-4x + 1}$	**12.** $6(x + 1) - 5(x + 2)$ $6x + 6 - 5x - 10$ = $\boxed{x - 4}$
13. $-5(k + 6) + 7(k - 4)$ $-5k - 30 + 7k - 28$ = $\boxed{2k - 58}$	**14.** $8(x + 5) - 4(x + 4)$ $8x + 40 - 4x - 16$ = $\boxed{4x + 24}$
15. $-9(m + 2) + 4(6 - 7m)$ $-9m - 18 + 24 - 28m$ = $\boxed{-37m + 6}$	**16.** $-3(-9v - 4) - 2(v - 2)$ $27v + 12 - 2v + 4$ = $\boxed{25v + 16}$
17. $-4(8 - c) - (3c + 7)$ $-32 + 4c - 3c - 7$ = $\boxed{c - 39}$	**18.** $6(p + 6) - (p - 6)$ $6p + 36 - p + 6$ = $\boxed{5p + 42}$
19. $21 - 8(w + 3) + 3 + 7w$ $21 - 8w - 24 + 3 + 7w$ = $\boxed{-w}$	**20.** $\frac{1}{2}\left(\frac{3}{5}x - 8\right) + \frac{7}{4}\left(\frac{2}{5}x + 4\right)$ $\frac{3}{10}x - 4 + \frac{7}{10}x + 7$ = $\boxed{x + 3}$

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Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples	
Key Words & Phrases	**ADDITION +** • sum • total • plus • increased by • more than	**SUBTRACTION −** • difference • minus • decreased by • less than * • subtracted from*
	MULTIPLICATION ● • times • product • of • twice • double / triple	**DIVISON /** • quotient • separate • divided by • half / third •
	(Parentheses Phrases) • times "the sum of" • times "the difference of / between"	**"Turn Around" Phrases** • less than • subtracted from
Examples	**Directions:** Translate each expression or equation.	
	1. "the product of a number and 3" $3n$	**2.** "the difference between a number and 10" $n-10$
	3. "the quotient of a number and -5" $\frac{n}{-5}$	**4.** "four subtracted from a number" $n-4$
	5. "the sum of twice a number and 7" $2n+7$	**6.** "the product of a number and 10, increased by 1" $10n+1$

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	7. "four less than a number squared" n^2-4		**8.** "twice a number divided by 5" $\frac{2n}{5}$	
Equations	**9.** "the sum of twice a number and 10 is 36" $2n+10=36$		**10.** "three times the difference of a number and 7 is 18" $3(n-7)=18$	
	11. "four times a number divided by 8 is 3" $\frac{4n}{8}=3$		**12.** "five times the sum of a number and 6 is 48" $5(n+6)=48$	
	13. "one half of a number is 8 less than the number itself" $\frac{1}{2}n = n-8$		**14.** "the product of a number and 3 is the same as the sum of that number and six" $3n = n+6$	
	15. "fifteen decreased by the quotient of a number and 3 is 20" $15-\frac{n}{3}=20$		**16.** "four less than the product of a number and 7 is eight more than that number" $7n-4 = 8+n$	
Inequalities	**LESS THAN**	**LESS THAN OR EQUAL TO**	**GREATER THAN**	**GREATER THAN OR EQUAL TO**
	$<$	$\leq$	$>$	$\geq$
Examples	**17.** "a number is less than -10" $n<-10$		**18.** "a number is less than or equal to 8" $n\leq 8$	
	19. "a number is greater than or equal to -4" $n\geq -4$		**20.** "a number is at most 45" $n\leq 45$	
	21. "a speed limit of 55 mph" $n\leq 55$		**22.** "a number is at least 16" $n\geq 16$	
	23. "a number is no less than 10" $n\geq 10$		**24.** "a number is no more than 26" $n\leq 26$	
	25. "a maximum number of 25" $n\leq 25$		**26.** "a minimum number of 5" $n\geq 5$	

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Name: _____ **Unit 1:** Algebra Basics

Date: _____ **Bell:** _____ **Homework 9:** Translating Expressions Equations, & Inequalities

Directions: Translate each expression.	
1. "the sum of a number and 4." $n+4$	**2.** "the quotient of a number and -2." $\frac{n}{-2}$
3. "the product of a number and 5." $5n$	**4.** "the difference of a number and 7." $n-7$
5. "a number squared increased by 1" n^2+1	**6.** "three less than twice a number" $2n-3$

Directions: Translate each equation. ***DO NOT SOLVE!***	
7. "twice a number divided by 6 is 42." $\frac{2n}{6}=42$	**8.** "two times the sum of number and 5 is 20" $2(n+5)=20$
9. "four times the difference of a number and 7 is 32" $4(n-7)=32$	**10.** "one-third of a number is 12 less than the number itself" $\frac{1}{3}n = n-12$
11. "ten increased by the quotient of a number and 2 is -15" $10+\frac{n}{2}=-15$	**12.** "one less than the product of a number and 3 is 5 more than the number itself" $3n-1=5+n$

Directions: Translate each inequality.	
13. "a number less than 15" $n<15$	**14.** "a number greater than or equal to -3" $n\geq -3$
15. "a number that is at least 90" $n\geq 90$	**16.** "a speed limit of 65 mph" $n\leq 65$
17. "a number that is no more than 20" $n\leq 20$	**18.** "a maximum number of 50" $n\leq 50$
19. "a minimum number of 10" $n\geq 10$	**20.** "a number no less than 25" $n\geq 25$

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WARM-UP!!!

Translate the following expressions, equations, & inequalities:

1. "The product of a number and 7" — $7n$
2. "Nine subtracted from twice a number" — $2n - 9$
3. "One less than the quotient of a number and -5." — $\frac{n}{-5} - 1$
4. "Three times the sum of a number and 10 is 24" — $3(n + 10) = 24$
5. "The difference of twice a number and 3 is -21." — $2n - 3 = -21$
6. "One-third of a number increased by 8 is 37." — $\frac{1}{3}n + 8 = 37$
7. "x is at most 6" — $x \leq 6$
8. "You must be at least 18 years old to vote" — $x \geq 18$

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Name: _____ **Algebra I**

Date: _____ **Bell:** _____ **Unit 1:** Algebra Basics

Quiz 1-3: Simplifying & Translating Expressions

Part I: Simplifying Expressions Simplifying by distributing and/or combining like terms.

1. $\frac{-2}{3}\left(\frac{5}{6}x - 3\right)$

2. $3c(5a + 2b - 1)$

3. $5x - 4 - 7x + 14$

4. $-3 - 5x - 5y + 8x + 7y + 8$

5. $4x^2y + 2xy - 6 + 5x^2y - 8xy + 13$

6. $2(x - 1) + 5(x + 9)$

$2x - 2 + 5x + 45$

7. $3(x + 9) - (x - 6)$

$3x + 27 - x + 6$

8. $3(5x + 9 - 4x)$

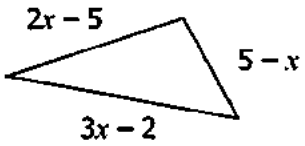
$15x + 27 - 12x$

9. $a + 3a - 4(9 - a)$

$a + 3a - 36 + 4a$

Answers	
1.	$\frac{-5}{9}x + 2$
2.	$15ac + 66c - 3c$
3.	$-2x + 10$
4.	$3x + 2y + 5$
5.	$9x^2y - 6xy + 7$
6.	$7x + 43$
7.	$2x + 33$
8.	$3x + 27$
9.	$8a - 36$
10.	$4x - 2$

10. Write the perimeter of the triangle as a simplified expression.



Part II: Translating Expressions, Equations, and Inequalities.

EXPRESSIONS:

11. "the product of a number and 4"

12. "five times the difference of a number and 3"

13. "seven less than a number squared"

Answers	
11.	$4n$
12.	$5(n - 3)$
13.	$n^2 - 7$

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EQUATIONS:

14. "the quotient of a number and 2, increased by 9 is -15"

15. "one-fourth of a number minus 8 is -23"

16. "the sum of twice a number and 6 is the same as 3 subtracted from the number itself"

INEQUALITIES

17. "a number greater than -5"

18. "a number that is at most 30"

19. "a minimum number of 17"

20. "a number that is at least 90"

Answers

14. $\frac{n}{2} + 9 = -15$
15. $\frac{1}{4}n - 8 = -23$
16. $2n + 6 = n - 3$
17. $n > -5$
18. $n \leq 30$
19. $n \geq 17$
20. $n \geq 90$

Part III: REVIEW! Circle the best answer.

21. Which number is an integer?

(A.) -18 **B.** $\frac{4}{7}$ **C.** $\sqrt{15}$ **D.** 0.35

22. Which set of real numbers does **NOT** contain the number 0?

A. Whole Numbers
(B.) Natural Numbers
C. Integers
D. Rational Numbers

23. Which property justifies the statement $(a - b) \cdot 0 = 0$?

A. Inverse Property
B. Associative Property
C. Distributive Property
(D.) Zero Product Property

24. Which property justifies the statement if $y = 7$, then $7 = y$?

A. Reflexive Property
B. Identity Property
(C.) Symmetric Property
D. Transitive Property

25. Which set of numbers is closed under subtraction?

A. Natural Numbers
(B.) Integers
C. Irrational Numbers
D. Odd Numbers

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Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples	
One-Step Equations	**1.** $m + 12 = 10$ $-12 \quad -12$ $m = -2$	**2.** $-2 = g - 9$ $+9 \quad +9$ $7 = g$
	3. $\frac{-7y}{-7} = \frac{-91}{-7}$ $y = 13$	**4.** $\frac{a}{9} = -4 \cdot 9$ $a = -36$
Fractions To "get rid" of a fraction, multiply by the reciprocal!	**5.** $\frac{3}{2} \cdot \frac{2}{3}x = 10 \cdot \frac{3}{2}$ $x = 15$	**6.** $\frac{9}{4} \cdot \frac{4}{9}w = -8 \cdot \frac{9}{4}$ $w = -18$
	7. $\frac{-5}{6} \cdot -\frac{6}{5}k = 12 \cdot \frac{-5}{6}$ $k = -10$	**8.** $-2 \cdot -\frac{1}{2}m = -9 \cdot -2$ $m = 18$
Two-Step Equations	**To Solve a Two-Step Equation:** 1. Undo the **Addition/Subtraction** (to remove constant term) 2. Undo the **Multiplication/Division** (to remove coefficient)	
	9. $6x + 8 = 50$ $-8 \quad -8$ $\frac{6x}{6} = \frac{42}{6}$ $x = 7$	**10.** $2n - 5 = 11$ $+5 \quad +5$ $\frac{2n}{2} = \frac{16}{2}$ $n = 8$
	11. $13 = -4k + 9$ $-9 \quad -9$ $\frac{4}{-4} = \frac{-4k}{-4}$ $k = -1$	**12.** $7 - 3y = 34$ $-7 \quad -7$ $\frac{-3y}{-3} = \frac{27}{-3}$ $y = -9$

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13. $\frac{x}{2} - 7 = 9$

$+7 \quad +7$

$2 \cdot \frac{x}{2} = 16 \cdot 2$

$\boxed{x = 32}$

14. $11 = \frac{c}{-5} + 8$

$-8 \quad -8$

$-5 \cdot 3 = \frac{c}{-5} \cdot -5$

$\boxed{-15 = c}$

15. $\frac{3}{5}x + 22 = 28$

$-22 \quad -22$

$\frac{5}{3} \cdot \frac{3}{5} x = 6 \cdot \frac{5}{3}$

$\boxed{x = 10}$

16. $-\frac{1}{3}m + 1 = -7$

$-1 \quad -1$

$-3 \cdot -\frac{1}{3}m = -8 \cdot -3$

$\boxed{m = 24}$

17. $-10 \div \frac{7}{4}p = -38$

$+10 \quad +10$

$\frac{4}{7} \cdot \frac{7}{4} p = -28 \cdot \frac{4}{7}$

$\boxed{p = -16}$

18. $15 = 9 - \frac{1}{2}x$

$-9 \quad -9$

$-2 \cdot 6 = -\frac{1}{2}x \cdot -2$

$\boxed{-12 = x}$

Watch Out!

The examples below are different in that the multiplication/division is done FIRST, followed by the addition/subtraction.

19. $\frac{x+11}{8} = -3 \cdot 8$

$x + 11 = -24$

$-11 \quad -11$

$\boxed{x = -35}$

20. $\frac{n-5}{-2} = -7 \cdot -2$

$n - 5 = 14$

$+5 \quad +5$

$\boxed{n = 19}$

21. $1 = \frac{a-13}{-6} \cdot -6$

$-6 = a - 13$

$+13 \quad +13$

$\boxed{7 = a}$

22. $4 = \frac{w+8}{9} \cdot 9$

$36 = w + 8$

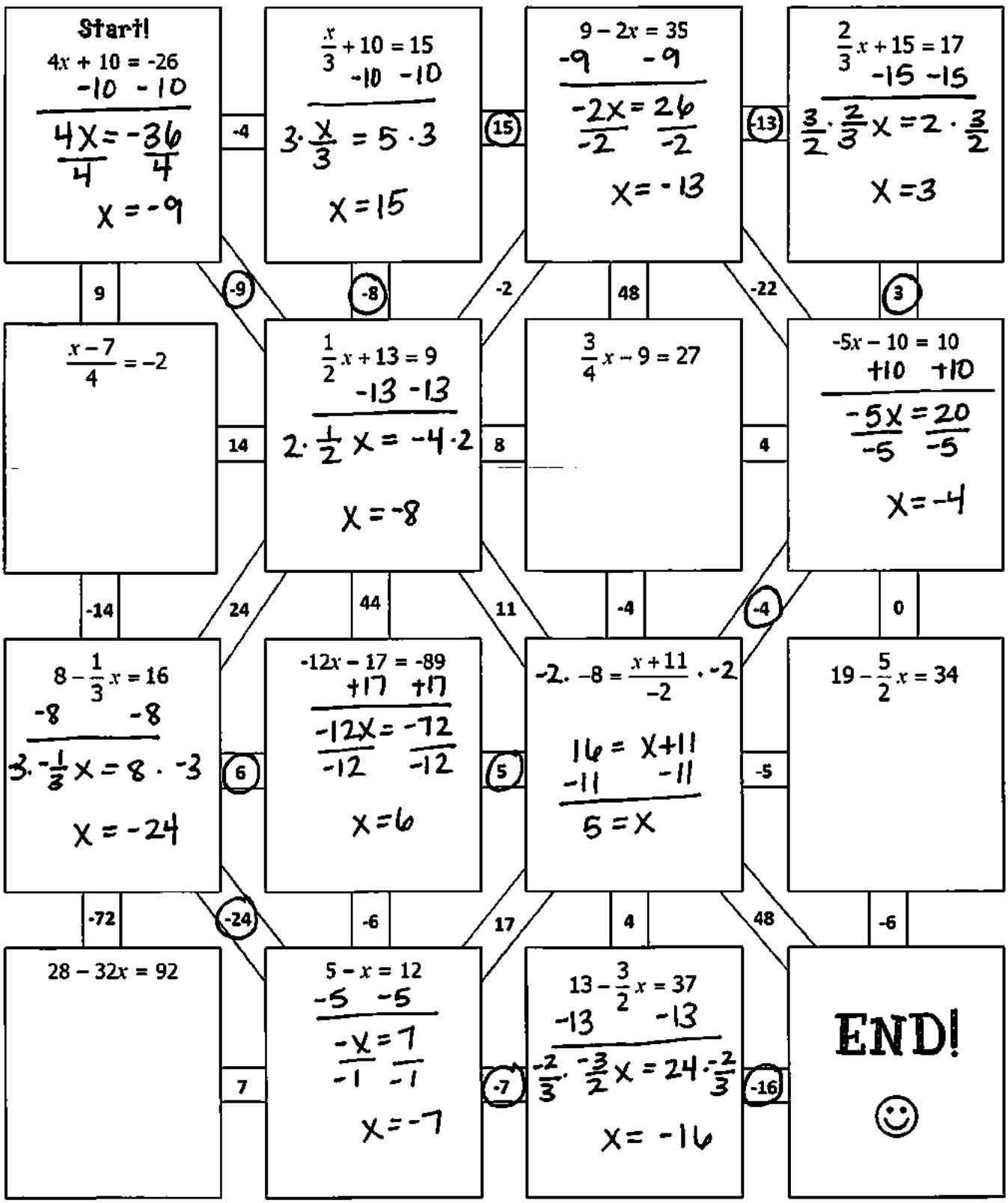
$-8 \quad -8$

$\boxed{28 = w}$

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Two-Step Equation Maze!

Directions: Use your solutions to navigate through the puzzle. **SHOW ALL STEPS!!!!**



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Name: _____ **Unit 1:** Algebra Basics

Date: _____ **Bell:** _____ **Homework 10:** One & Two-Step Equations

Directions: Solve each equation. **SHOW ALL STEPS!**

1. $y - 17 = -37$

$+17 \quad +17$

$y = -20$

2. $-12 + k = 18$

$+12 \quad +12$

$K = 30$

3. $5m = -20$

$\frac{5m}{5} = \frac{-20}{5}$

$m = -4$

4. $\frac{x}{9} = 3$ $\cdot 9$

$X = 27$

5. $\frac{4}{5}z = 16$ $\cdot \frac{5}{4}$

$Z = 20$

6. $3p - 4 = 11$

$+4 \quad +4$

$\frac{3p}{3} = \frac{15}{3}$

$p = 5$

7. $-5a + 7 = -38$

$-7 \quad -7$

$\frac{-5a}{-5} = \frac{-45}{-5}$

$a = 9$

8. $31 - 12n = 211$

$-31 \quad -31$

$\frac{-12n}{-12} = \frac{180}{-12}$

$n = -15$

9. $\frac{x}{3} - 15 = -7$

$+15 \quad +15$

$3 \cdot \frac{x}{3} = 8 \cdot 3$

$X = 24$

10. $\frac{f}{-4} + 11 = 5$

$-11 \quad -11$

$-4 \cdot \frac{f}{-4} = -6 \cdot -4$

$f = 24$

11. $\frac{4}{3}v + 25 = 33$

$-25 \quad -25$

$\frac{3}{4} \cdot \frac{4}{3}v = 8 \cdot \frac{3}{4}$

$v = 6$

12. $\frac{5}{8}c + 13 = 18$

$-13 \quad -13$

$\frac{8}{5} \cdot \frac{5}{8}c = 5 \cdot \frac{8}{5}$

$C = 8$

13. $-\frac{2}{3}w + 53 = 59$

$-53 \quad -53$

$-\frac{3}{2} \cdot -\frac{2}{3}w = 6 \cdot -\frac{3}{2}$

$w = -9$

14. $11 = \frac{4}{5}h - 1$

$+1 \quad +1$

$\frac{5}{4} \cdot 12 = \frac{4}{5}h \cdot \frac{5}{4}$

$15 = h$

15. $-\frac{3}{8}r + 41 = -22$

$-41 \quad -41$

$-\frac{8}{3} \cdot -\frac{3}{8}r = -63 \cdot -\frac{8}{3}$

$r = 168$

16. $34 = 28 - \frac{2}{5}y$

$-28 \quad -28$

$-\frac{5}{2} \cdot 6 = -\frac{2}{5}y \cdot -\frac{5}{2}$

$-15 = y$

17. $\frac{n+4}{-2} = -9$ $\cdot -2$

$n + 4 = 18$

$-4 \quad -4$

$n = 14$

18. $-5 = \frac{x-1}{8}$ $\cdot 8$

$-40 = X - 1$

$+1 \quad +1$

$-39 = X$

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Name:	Date:
Topic:	Class:

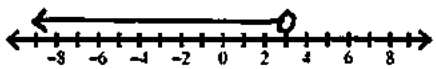
Main Ideas/Questions	Notes/Examples

Rules for GRAPHING INEQUALITIES

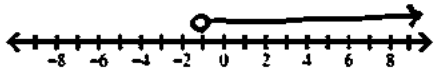
- ✓ Use an open circle for < or > signs.
- ✓ Use a closed circle for ≤ or ≥ signs.

Graph the following inequalities on the number line.

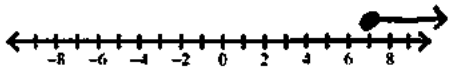
1. $x < 3$



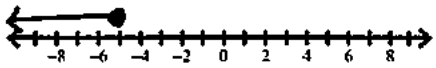
2. $x > -1$



3. $x \geq 7$

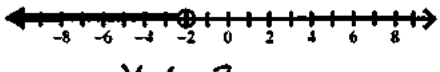


4. $x \leq -5$



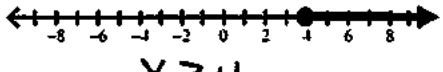
Write an equality given the graph.

5.



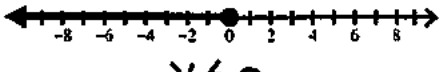
$x < -2$

6.



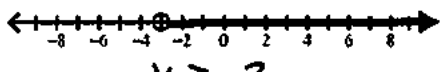
$x \geq 4$

7.



$x \leq 0$

8.



$x > -3$

Rules for SOLVING INEQUALITIES

You must flip the inequality symbol when you multiply or divide by a negative number!

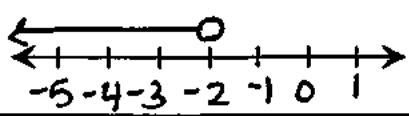
Solve and graph the following inequalities.

9. $3x + 2 < -4$

$-2 \quad -2$

$\frac{3x}{3} < \frac{-6}{3}$

$x < -2$

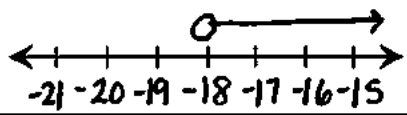


10. $\frac{x}{6} + 2 > -1$

$-2 \quad -2$

$6 \cdot \frac{x}{6} > -3 \cdot 6$

$x > -18$



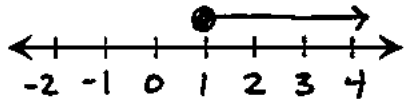
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11. $-7x + 8 \le 1$

$-8 \quad -8$

$\frac{-7x}{-7} \le \frac{-7}{-7}$

$x \ge 1$

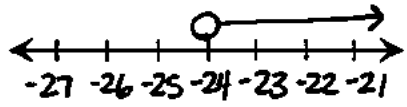


12. $\frac{x}{-8} - 5 < -2$

$+5 \quad +5$

$-8 \cdot \frac{x}{-8} < 3 \cdot -8$

$x > -24$

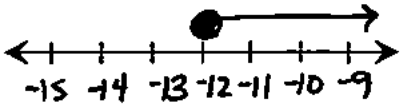


13. $-2x - 5 \le 19$

$+5 \quad +5$

$\frac{-2x}{-2} \le \frac{24}{-2}$

$x \ge -12$

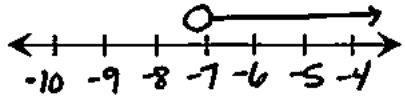


14. $3 - 4x < 31$

$-3 \quad -3$

$\frac{-4x}{-4} < \frac{28}{-4}$

$x > -7$

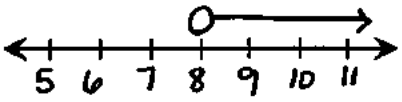


15. $\frac{3}{4}x + 5 > 11$

$-5 \quad -5$

$\frac{4}{3} \cdot \frac{3}{4}x > 6 \cdot \frac{4}{3}$

$x > 8$

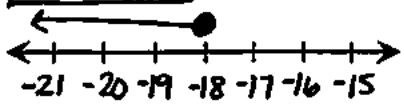


16. $-\frac{5}{6}x - 1 \ge 14$

$+1 \quad +1$

$-\frac{6}{5} \cdot -\frac{5}{6}x \ge 15 \cdot -\frac{6}{5}$

$x \le -18$

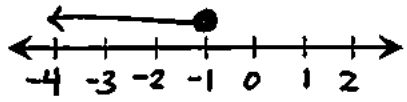


17. $5x - 2 \le -7$

$+2 \quad +2$

$\frac{5x}{5} \le \frac{-5}{5}$

$x \le -1$

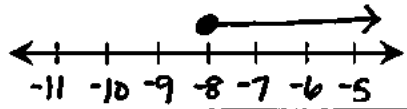


18. $6 - x \le 14$

$-6 \quad -6$

$\frac{-x}{-1} \le \frac{8}{-1}$

$x \ge -8$

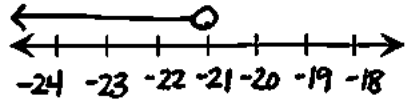


19. $4 - \frac{1}{3}x > 11$

$-4 \quad -4$

$-3 \cdot -\frac{1}{3}x > 7 \cdot -3$

$x < -21$

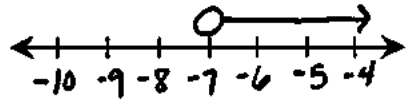


20. $\frac{x+11}{-2} < -2 \cdot -2$

$x + 11 > 4$

$-11 \quad -11$

$x > -7$



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Name: _____ **Unit 1:** Algebra Basics

Date: _____ **Bell:** _____ **Homework 11:** Solving & Graphing Inequalities

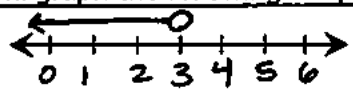
Directions: Solve and graph the following inequalities. **SHOW ALL STEPS!**

1. $3x + 1 < 10$

$-1 \quad -1$

$\frac{3x}{3} < \frac{9}{3}$

$x < 3$

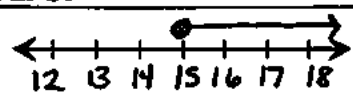


2. $\frac{x}{3} + 10 \geq 15$

$-10 \quad -10$

$3 \cdot \frac{x}{3} \geq 5 \cdot 3$

$x \geq 15$

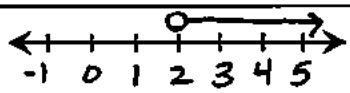


3. $-2x + 22 < 18$

$-22 \quad -22$

$\frac{-2x}{-2} < \frac{-4}{-2}$

$x > 2$

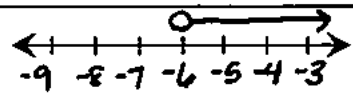


4. $7x + 11 > -31$

$-11 \quad -11$

$\frac{7x}{7} > \frac{-42}{7}$

$x > -6$

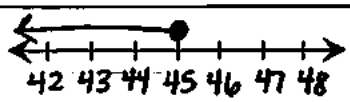


5. $\frac{x}{5} - 3 \leq 6$

$+3 \quad +3$

$5 \cdot \frac{x}{5} \leq 9 \cdot 5$

$x \leq 45$

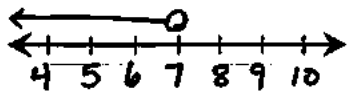


6. $18 - 2x > 4$

$-18 \quad -18$

$\frac{-2x}{-2} > \frac{-14}{-2}$

$x < 7$

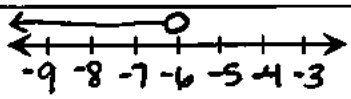


7. $\frac{x}{-2} + 1 > 4$

$-1 \quad -1$

$-2 \cdot \frac{x}{-2} > 3 \cdot -2$

$x < -6$

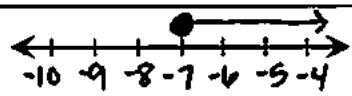


8. $4 - \frac{x}{7} \leq 5$

$-4 \quad -4$

$-7 \cdot \frac{-x}{7} \leq 1 \cdot -7$

$x \geq -7$

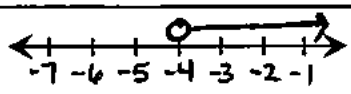


9. $-3x - 4 < 8$

$+4 \quad +4$

$\frac{-3x}{-3} < \frac{12}{-3}$

$x > -4$

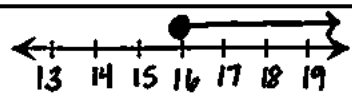


10. $\frac{5}{8}x - 1 \geq 9$

$+1 \quad +1$

$\frac{8}{5} \cdot \frac{5}{8}x \geq 10 \cdot \frac{8}{5}$

$x \geq 16$

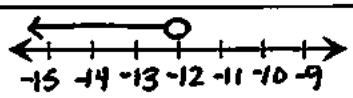


11. $-\frac{3}{4}x + 2 > 11$

$-2 \quad -2$

$-\frac{4}{3} \cdot \frac{-3}{4}x > 9 \cdot -\frac{4}{3}$

$x < -12$

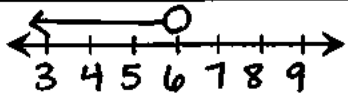


12. $\frac{x+15}{-3} > -7 \cdot -3$

$x + 15 < 21$

$-15 \quad -15$

$x < 6$



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Translate... and Solve!

1) "Twice a number plus 4 is 8"	$2n+4=8$ $-4\ \ -4$ $\frac{2n}{2}=\frac{4}{2}$ $\boxed{n=2}$	2) "The product of 4 and a number plus 17 is 5."	$4n+17=5$ $-17\ \ -17$ $\frac{4n}{4}=\frac{-12}{4}$ $\boxed{n=-3}$
3) "Twice a number minus 5 is 7"	$2n-5=7$ $+5\ \ +5$ $\frac{2n}{2}=\frac{12}{2}$ $\boxed{n=6}$	4) "The quotient of a number and 6 minus 3 is 1"	$\frac{n}{6}-3=1$ $+3\ \ +3$ $6\cdot\frac{n}{6}=4\cdot 6$ $\boxed{n=24}$
5) "Two more than 4 times a number is -18"	$4n+2=-18$ $-2\ \ -2$ $\frac{4n}{4}=\frac{-20}{4}$ $\boxed{n=-5}$	6) "Seven less than twice a number is 5"	$2n-7=5$ $+7\ \ +7$ $\frac{2n}{2}=\frac{12}{2}$ $\boxed{n=6}$
7) "Five less than the quotient of a number and 3 is -7"	$\frac{n}{3}-5=-7$ $+5\ \ +5$ $3\cdot\frac{n}{3}=-2\cdot 3$ $\boxed{n=-6}$	8) "One subtracted from the product of 4 and a number is 11"	$4n-1=11$ $+1\ \ +1$ $\frac{4n}{4}=\frac{12}{4}$ $\boxed{n=3}$
9) "The quotient of a number and -9, increased by 10 is 11"	$\frac{n}{-9}+10=11$ $-10\ \ -10$ $\frac{n}{-9}=1\cdot -9$ $\boxed{n=-9}$	10) "Half of a number decreased by 8 is -3"	$\frac{1}{2}n-8=-3$ $+8\ \ +8$ $2\cdot\frac{1}{2}n=5\cdot 2$ $\boxed{n=10}$

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11) "The sum of three-fourths of a number and 1 is -5"	$\frac{3}{4}n + 1 = -5$ $-1 \quad -1$ $\frac{4}{3}\cdot\frac{3}{4}n = -6 \cdot \frac{4}{3}$ $\boxed{n = -8}$	12) "Ten minus one-third of a number is 4"	$10 - \frac{1}{3}n = 4$ $-10 \quad -10$ $-3 \cdot -\frac{1}{3}n = -6 \cdot -3$ $\boxed{n = 18}$
13) "Four more than the quotient of a number and 3 is at least 9"	$4 + \frac{n}{3} \geq 9$ $-4 \quad -4$ $3 \cdot \frac{n}{3} \geq 5 \cdot 3$ $\boxed{n \geq 15}$	14) "Negative three times a number increased by seven is less than -11"	$-3n + 7 < -11$ $-7 \quad -7$ $\frac{-3n}{-3} < \frac{-18}{-3}$ $\boxed{n > 6}$
15) "Seven more than one-sixth of a number is greater than three"	$7 + \frac{1}{6}n > 3$ $-7 \quad -7$ $6 \cdot \frac{1}{6}n > -4 \cdot 6$ $\boxed{n > -24}$	16) "The sum of 5 and the twice a number is at most 27"	$5 + 2n \leq 27$ $-5 \quad -5$ $\frac{2n}{2} \leq \frac{22}{2}$ $\boxed{n \leq 11}$
17) "Four less than two-fifths of a number is no less than 2"	$\frac{2}{5}n - 4 \geq 2$ $+4 \quad +4$ $\frac{5}{2}\cdot\frac{2}{5}n \geq 6 \cdot \frac{5}{2}$ $\boxed{n \geq 15}$	18) "Three-fourths of a number decreased by 10 is greater than or equal 5"	$\frac{3}{4}n - 10 \geq 5$ $+10 \quad +10$ $\frac{4}{3}\cdot\frac{3}{4}n \geq 15 \cdot \frac{4}{3}$ $\boxed{n \geq 20}$
19) "The quotient of a number and -5, increased by 7 is at minimum 1"	$\frac{n}{-5} + 7 \geq 1$ $-7 \quad -7$ $-5 \cdot \frac{n}{-5} \geq -6 \cdot -5$ $\boxed{n \leq 30}$	20) "One less than half a number is a maximum of 19"	$\frac{1}{2}n - 1 \leq 19$ $+1 \quad +1$ $2\frac{1}{2}n \leq 20 \cdot 2$ $\boxed{n \leq 40}$

GOOD JOB!

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Unit 1 Test Study Guide
Algebra Basics

Name: _____

Date: _____ Block: _____

Topic 1: The Real Number System

List ALL sets to which each number belongs. (Use *R, I, Q, Z, W, N*)		
1. $-\frac{14}{2}$ Z, Q, R	2. $\sqrt{64}$ N, W, Z, Q, R	3. 0 W, Z, Q, R
4. π I, R	5. $0.\overline{45}$ Q, R	6. $\frac{3}{8}$ Q, R

Place the LETTER of each value its location in the real number system below.

A. $-0.\overline{2}$ B. 18

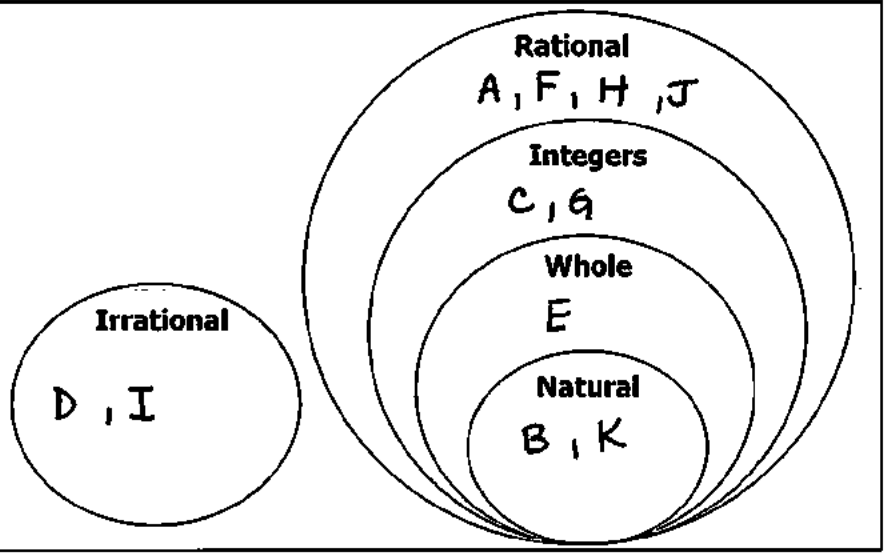
C. $-\sqrt{100}$ D. π

E. 0 F. $2\frac{1}{6}$

G. -5 H. 4.03

I. $-\sqrt{72}$ J. $\sqrt{\frac{4}{9}}$

K. $\frac{36}{9}$



Topic 2: Properties

Identify the property shown below.	
7. $4 + (x + y) = (4 + x) + y$ Associative	8. $\frac{2}{5} \cdot \frac{5}{2} = 1$ Inverse
9. if $\sqrt{49} = 7$ and $7 = 3 + 4$, then $\sqrt{49} = 3 + 4$ Transitive	10. $-28 = -28$ Reflexive
11. $8x^2 \cdot 1 = 8x^2$ Identity	12. $10y + (-10y) = 0$ Inverse
13. $(a + 4) \cdot 0 = 0$ Zero Product	14. $-5(x + 7) = -5x - 35$ Distributive
15. $(x + 2) + y = (2 + x) + y$ Commutative	16. if $x = -1$, then $-1 = x$ Symmetric

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Answer true or false. Provide an example if false.

17. The set of negative numbers are closed under division.

False; $\frac{-36}{-9} = 4$

18. The set of whole numbers are closed under subtraction.

False; $0 - 6 = -6$

19. The set of integers are closed under multiplication.

True

Topic 3: Absolute Value & Order of Operations

20. $|-21| + |5|$

$21 + 5 = \boxed{26}$

21. $|-10 - 4|$

$|-14| = \boxed{14}$

22. $10 - 2 \cdot 3^2 + 13$

$10 - 2 \cdot 9 + 13$

$10 - 18 + 13$

$-8 + 13 = \boxed{5}$

23. $[(-5+1) \div 2]^3 - |-7|$

$[(-4) \div 2]^3 - 7$

$[-2]^3 - 7$

$-8 - 7 = \boxed{-15}$

24. $\frac{25 + 5^2 \div 5}{7 - 12 \div 4 + 2}$

$\frac{25 + 25 \div 5}{7 - 3 + 2} = \frac{25 + 5}{4 + 2} = \frac{30}{6} = \boxed{5}$

25. $\frac{(3-7)^2 + 11}{|-2| + |-1|}$

$\frac{(-4)^2 + 11}{2 + 1} = \frac{16 + 11}{3} = \frac{27}{3} = \boxed{9}$

Topic 4: Evaluating Expressions

26. $3x^3 - 8y^2$ if $x = 2$ and $y = -3$

$3(2)^3 - 8(-3)^2$

$3(8) - 8(9)$

$24 - 72 = \boxed{-48}$

27. $|a - 4b|$ if $a = 7$ and $b = 2$

$|7 - 4(2)|$

$|7 - 8| = |-1| = \boxed{1}$

28. $w^2 + 2xy$ if $x = -3$, $w = -2$, and $y = 1$

$(-2)^2 + 2(-3)(1)$

$4 - 6(1)$

$4 - 6 = \boxed{-2}$

29. $\frac{7c^2 + 5}{4a - b}$ if $a = 1$, $b = -5$ and $c = -4$

$\frac{7(-4)^2 + 5}{4(1) - (-5)}$ $\frac{7(16) + 5}{4 + 5} = \frac{117}{9} = \boxed{13}$

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Topic 5: Translating Expressions, Equations, & Inequalities

30. "The product of a number and 7, increased by three."

$7n+3$

31. "One less than twice a number."

$2n-1$

32. "Four times the difference of a number and nine is -30."

$4(n-9) = -30$

33. "Five more than the quotient of a number and eight is 42."

$5 + \frac{n}{8} = 42$

34. "The schola can hold a maximum of 150 people."

$a \leq 150$

35. "You must be at least 25 to rent a car."

$a \geq 25$

Topic 6: Simplifying Expressions

36. $n + 4 - 9 - 5n$

$-4n - 5$

37. $7x - 4y - 8 - 3x - 8y + 12$

$4x - 12y + 4$

38. $18 - 2(4x + 7) + 5x$

$18 - 8x - 14 + 5x$

$= -3x + 4$

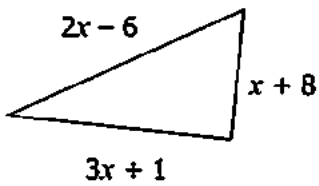
39. $9(x - 3) - (x + 2)$

$9x - 27 - x - 2$

$= 8x - 29$

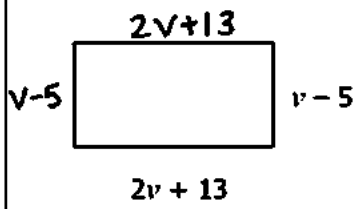
40. Write the perimeter of the triangle in simplest form.

$6x + 3$



41. Write the perimeter of the rectangle in simplest form.

$6v + 16$



Topic 7: Two-Step Equations

42. $3x - 7 = 11$

$+7 \quad +7$

$\frac{3x}{3} = \frac{18}{3}$

$x = 6$

43. $\frac{x}{4} + 8 = 3$

$-8 \quad -8$

$4 \cdot \frac{x}{4} = -5 \cdot 4$

$x = -20$

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44. $\frac{2}{5}x - 3 = -17$

$+3 \quad +3$

$\frac{5}{2} \cdot \frac{2}{5}x = -14 \cdot \frac{5}{2}$

$\boxed{x = -35}$

45. $9 - 7x = 16$

$-9 \quad -9$

$\frac{-7x}{-7} = \frac{7}{-7}$

$\boxed{x = -1}$

46. $9 - \frac{3}{4}x = 57$

$-9 \quad -9$

$-\frac{4}{3} \cdot -\frac{3}{4}x = 48 \cdot -\frac{4}{3}$

$\boxed{x = -64}$

47. $\frac{x-17}{3} = -3 \cdot 3$

$x - 17 = -9$

$+17 \quad +17$

$\boxed{x = 8}$

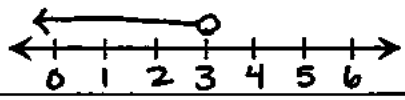
Topic 8: Solving & Graphing Inequalities

48. $3x + 5 < 14$

$-5 \quad -5$

$\frac{3x}{3} < \frac{9}{3}$

$\boxed{x < 3}$

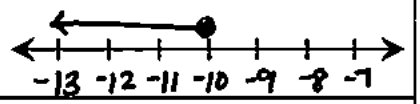


49. $-5x + 2 \geq 52$

$-2 \quad -2$

$\frac{-5x}{-5} \geq \frac{50}{-5}$

$\boxed{x \leq -10}$
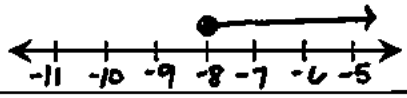


50. $\frac{x}{-2} + 11 \leq 15$

$-11 \quad -11$

$-2 \cdot \frac{x}{-2} \leq 4 \cdot -2$

$\boxed{x \geq -8}$

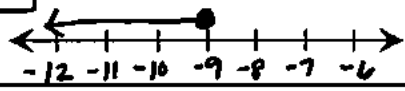


51. $4x - 1 \leq -37$

$+1 \quad +1$

$\frac{4x}{4} \leq \frac{-36}{4}$

$\boxed{x \leq -9}$

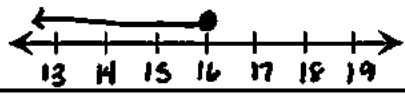


52. $\frac{5}{8}x - 1 \leq 9$

$+1 \quad +1$

$\frac{8}{5} \cdot \frac{5}{8}x \leq 10 \cdot \frac{8}{5}$

$\boxed{x \leq 16}$

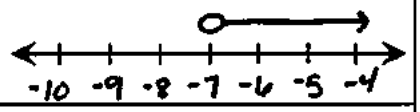


53. $4 - \frac{x}{7} < 5$

$-4 \quad -4$

$-7 \cdot -\frac{x}{7} < 1 \cdot -7$

$\boxed{x > -7}$
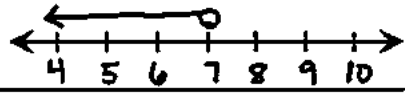


54. $18 - 2x > 4$

$-18 \quad -18$

$\frac{-2x}{-2} > \frac{-14}{-2}$

$\boxed{x < 7}$

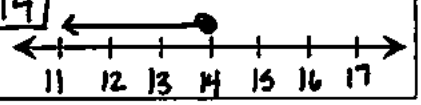


55. $8 - \frac{3}{2}x \geq -13$

$-8 \quad -8$

$-\frac{2}{3} \cdot -\frac{3}{2}x \geq -21 \cdot -\frac{2}{3}$

$\boxed{x \leq 14}$



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Name: _____

Date: _____ Bell: _____

Algebra I Unit 1 Test

(Algebra Basics)

SHOW ALL WORK NEEDED TO ANSWER EACH QUESTION!
PLACE YOUR FINAL ANSWER IN THE BOX. GOOD LUCK! ☺

1. Which is the smallest set of real numbers that contains the value below?

$$-\frac{18}{6}$$

A. Irrational Numbers
B. Rational Numbers
C. Natural Numbers
D. Integers

D

2. Which set contains the value below?

$$\sqrt{50}$$

A. Irrational Numbers
B. Natural Numbers
C. Rational Numbers
D. Integers

A

3. The set below only contains which types of numbers?

$$\{-1, 5, \frac{1}{2}, 15, 3.75, 36, \sqrt{81}, 100\}$$

A. Irrational Numbers
B. Rational Numbers
C. Integers
D. Natural Numbers

B

4. Select all sets to which the value below belongs.

$$\sqrt{2} - \sqrt{2}$$

- ☑ Real Numbers
- ☐ Irrational Numbers
- ☑ Rational Numbers
- ☑ Integers
- ☑ Whole Numbers
- ☐ Natural Numbers

5. Which of the following is *true* regarding number sets?

A. All integers are whole numbers.
B. All irrational numbers are real numbers.
C. All real numbers are integers.
D. All rational numbers are natural numbers.

B

6. Which property justifies the statement below?

$$x(y - 3) = xy - 3x$$

A. Associative Property
B. Transitive Property
C. Distributive Property
D. Commutative Property

C

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7. Which property justifies the statement below?

$$(a + b) + 0 = (a + b)$$

A. Reflexive Property
B. Identity Property
C. Symmetric Property
D. Associative Property

B

8. Which property justifies the statement below?

if $9x = 27$, then $27 = 9x$

A. Commutative property
B. Identity Property
C. Symmetric Property
D. Transitive Property

C

9. Give an example of the reflexive property.

*Answers may vary.

$5x = 5x$

10. Give an example of the associative property.

*Answers may vary.

$(5 + 2) + 7 = 5 + (2 + 7)$

11. Integers are *not* closed under which operation?

A. Addition
B. Subtraction
C. Multiplication
D. Division

D

12. Which of the following is a *false* statement?

A. Natural numbers are closed under division.
B. Integers are closed under subtraction.
C. Even numbers are closed under multiplication.
D. Real numbers are closed under addition.

A

13. Simplify the expression below.

$4 + 2^3 - |-4|$

$4 + 8 - 4$

$12 - 4$

8

14. Simplify the expression below.

$3^2 - 3 \cdot [(5^2 - 9) \div 2]$

$9 - 3[(25 - 9) \div 2]$

$9 - 3[16 \div 2]$

$9 - 3[8]$

$9 - 24$

-15

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15. Simplify the expression below

$$\frac{(3-13)^2+14}{4^2-5\cdot 2}$$

$$\frac{(-10)^2+14}{16-10}$$

$$\frac{100+14}{6}$$

$$\frac{114}{6}$$

19

16. Evaluate the expression below if $a = -8$, $b = 17$, and $c = 21$

$$a^2-(b+c)$$

$$(-8)^2-(17+21)$$

$$64-38$$

26

17. Evaluate the expression below if $x = -1$ and $y = 3$

$$3x^2-y^2$$

$$3(-1)^2-(3)^2$$

$$3(1)-9$$

$$3-9$$

-6

18. Evaluate the expression below if $a = 8$ and $b = -5$

$$|a|-|b|$$

$$|8|-|-5|$$

$$8-5$$

3

19. Translate statement below.

"One less than twice a number."

A. $1-n^2$

B. n^2-1

C. $1-2n$

D. $2n-1$

D

20. Translate the statement below.

"Five times the difference of a number and 3 is 17."

A. $5n-3=17$

B. $5(n-3)=17$

C. $\frac{5n}{3}=17$

D. $5\left(\frac{n}{3}\right)=17$

B

21. Translate the statement below.

"A number is no more than 50"

A. $x < 50$

B. $x \le 50$

C. $x > 50$

D. $x \ge 50$

B

22. Translate the statement below

"Your grade must be at least 64 to pass this class"

A. $g < 64$

B. $g \le 64$

C. $g > 64$

D. $g \ge 64$

D

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23. Simplify the expression below.

$$2m - 16 + 5m + 45$$

(A.) $7m + 29$
B. $7m + 61$
C. $-3m + 29$
D. $-3m + 61$

A

24. Simplify the expression below.

$$2x - 4y + 6 + 3x - 9y - 4$$

A. $5x + 13y + 2$
(B.) $5x - 13y + 2$
C. $5x - 5y + 2$
D. $-8xy + 2$

B

25. Simplify the expression below completely.

$$10 - 4(2x + 7)$$

$10 - 8X - 28$

$-8X - 18$

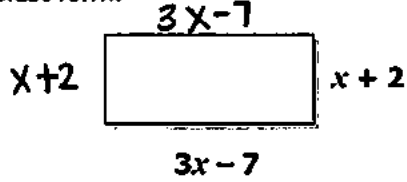
26. Simplify the expression below completely.

$$4(5w - 3) - (w - 1)$$

$20w - 12 - w + 1$

$19w - 11$

27. Give the perimeter of the rectangle below in simplest form.



$8X - 10$

28. Identify the first step to solve the equation below.

$$\frac{x}{-4} + 7 = -1$$

A. Add 4
(B.) Subtract 7
C. Multiply by -4
D. Add 1

B

29. Solve the equation below.

$$5x + 1 = -49$$
$$-1 \quad -1$$
$$\frac{5x}{5} = \frac{-50}{5}$$

$X = -10$

30. Solve the equation below.

$$\frac{2}{3}x - 5 = 7$$
$$+5 \quad +5$$
$$\frac{3}{2} \cdot \frac{2}{3}x = 12 \cdot \frac{3}{2}$$

$X = 18$

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31. Solve the equation below.

$$2 - 4a = 18$$
$$-2 \quad -2$$
$$-4a = 16$$
$$\frac{-4a}{-4} = \frac{16}{-4}$$

$a = -4$

32. Solve the equation below.

$$12 - \frac{1}{9}n = 17$$
$$-12 \quad -12$$
$$-9 \cdot -\frac{1}{9}n = 5 \cdot -9$$

$n = -45$

33. Solve the inequality below.

$$4x - 1 \leq -29$$
$$+1 \quad +1$$
$$\frac{4x}{4} \leq \frac{-28}{4}$$

$x \leq -7$

34. Solve the inequality below.

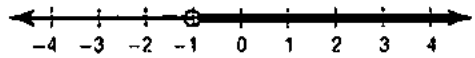
$$-6 \cdot \frac{x - 13}{-6} < 3 \cdot -6$$
$$x - 13 > -18$$
$$+13 \quad +13$$

$x > -5$

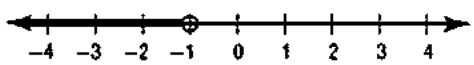
35. Which graph represents the solution to the inequality below?

$$4 - 2x > -2$$
$$-4 \quad -4$$
$$\frac{-2x}{-2} > \frac{-6}{-2}$$
$$x < 3$$

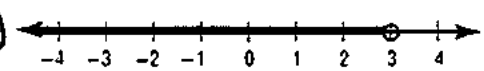
A.



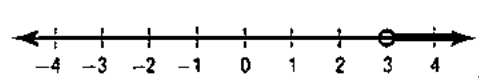
B.



C.

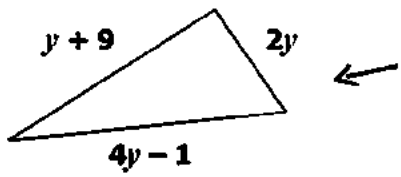


D.



C

B. Rob would like to create a triangle that is seven less than twice the perimeter of the triangle below. Give an expression for the perimeter of the new triangle in simplest form.



$7y + 8$

New Δ:

$2(7y + 8) - 7$

$14y + 16 - 7$

$14y + 9$

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